eCTD Compilation & Audit Platform — Complete Explainer

Sep 30, 2026 · @hil

This project is an AI system that assembles and cross-checks the manufacturing section (CMC, eCTD Module 3) of a drug approval filing, so that every number in the dossier is pulled from raw lab data, verified by code, traced to its exact source location, and signed by a human. Where the detailed overview is wrong or overclaims, this doc says so and gives the fix — look for the **Check:** notes.

Scope: the project started with CMC (Module 3) and now covers the whole eCTD, Modules 1 to 5. Module 3 is the first engine to build and the worked example throughout; the same design extends to every module.

How to read this doc: top to bottom. Each part builds on the one before it.

Part	Sections	What you will understand
1. The big picture	1	What the project is, on one page
2. The world and the problem	2–5	Drug approval, the eCTD and CMC; today's manual workflow; why existing tools fail; competitors
3. The core idea	6–9	Facts first; widening it to every claim; the evidence graph; the whole platform at a glance
4. The first engine: Module 3	10–20	Each component, why it was chosen over the alternatives, and one request end to end
5. Every other module	21–26	Modules 4, 5, 2 and 1, labeling, publishing, after submission, rules for patient data
6. Technology and timing	27–29	Why now, the latest GenAI and agent technologies, coming data standards
7. Risks, plan and business	30–34	Technical challenges, roadmap, market, how this differs from Weave Bio, product experience and design
8. Review and build	35–36	Every error in the original overview, and the student-scale build

Part	Sections	What you will understand
Appendix	37–38	Glossary, sources

1. The project on one page

Before a company may test a drug in people or sell it, it must prove to regulators that it can **make** that drug the same way, batch after batch. That proof is a dossier of thousands of numbers (eCTD Module 3). Today humans copy those numbers by hand from lab systems and scanned PDFs into Word, and one typo can stall an approval. This project builds that dossier automatically and audits it, and a human still signs.

Scope: the platform covers every module of the eCTD. It starts with Module 3 (CMC), the most number-heavy part, and the same design then extends to animal studies (Module 4), clinical studies (Module 5), the summaries (Module 2), the label and the regional Module 1 (sections 7–9 and 21–26).

Taking Module 3, the first engine, as the example:

What goes in	What comes out
Stability results from LIMS, chromatography results from Empower, lab notes from the ELN, batch records from SAP, scanned Certificates of Analysis from contract manufacturers	Draft Module 3 sections (for example 3.2.P.5 Control of Drug Product, 3.2.P.8 Stability) in which every number links to the exact cell it came from, a verification report, and a tamper-evident audit trail

Who uses it: regulatory medical writers (drafting), QA auditors (checking), and the regulatory lead (signing).

The single idea everything serves: a language model is good at reading messy documents and writing scientific prose, and bad at being the source of truth for a number. So the system splits the work:

1. Extraction turns every source into **facts** (value + unit + batch + test + condition + time point + exact page and box it came from).
2. **Code**, not the model, does every calculation (means, trends, limits).
3. The model **writes prose around verified facts**; it never types a digit itself.
4. A checker rejects any number in the draft that does not match a stored fact.

5. A **human** reviews every number against its source and signs; every step is logged.

The five building blocks (the overview's names in brackets):

1. Connectors that pull data from company systems [MCP servers].
2. Parsers that turn scanned PDFs into tables with coordinates [vision-language pipeline].
3. A knowledge store that remembers how batches, tests, conditions and results relate, and searches regulations [Neo4j + Pinecone].
4. A coordinator that runs the steps and the specialist agents [LangGraph supervisor].
5. A review screen and an audit ledger [provenance workspace + Part 11 ledger].

What it is not: it does not make regulatory decisions, does not file with FDA itself (it hands finished sections to Veeva Vault, which publishes and submits), and does not replace QA sign-off.

Verdict on the two overview docs, in short: the problem analysis is strong and mostly accurate. The architecture points the right way but is overloaded with brand names, contradicts itself on data sovereignty, names at least one component that does not exist, and contains a few pharmaceutical science errors. Section 35 lists every one with its fix.

2. The world it lives in

To get a drug approved, a company must prove three things: it works, it is safe, and it can be **made with consistent quality**. The third is called CMC (Chemistry, Manufacturing and Controls), and it is the only part this project touches.

The path of a drug and where CMC is filed:

1. Discovery and animal studies.
2. **IND** (Investigational New Drug application): permission to start human trials. It already needs a CMC section, smaller in early phases.
3. Phase 1, 2, 3 clinical trials. The CMC section grows as manufacturing scales up.
4. **NDA** (New Drug Application, for chemically made "small molecule" drugs) or **BLA** (Biologics License Application, for drugs made in living cells, such as antibodies and vaccines): permission to sell. The full CMC dossier goes in here.
5. **Post-approval changes**: every later change to a site, process, supplier or test method needs a supplement with updated CMC data.

Small molecule vs biologic matters for this project, because different rules apply. A small molecule (aspirin-sized, made by chemical synthesis) follows ICH Q3A/Q3B for impurities. A biologic (a protein roughly 1,000 times heavier, grown in a bioreactor) follows ICH Q6B and Q5C instead. The overview mixes the two; section 20 shows where.

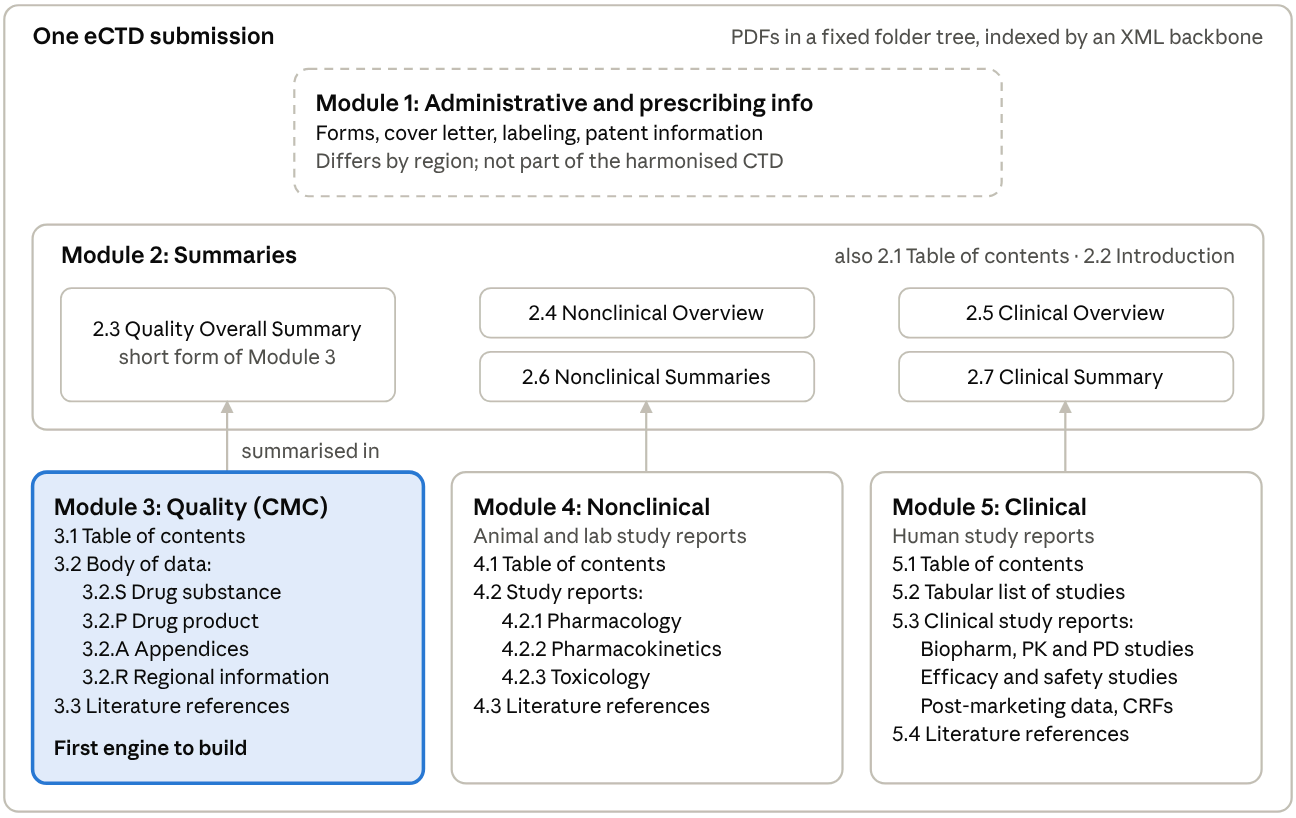
The eCTD (electronic Common Technical Document) is the standard format FDA, EMA, Japan's PMDA and others accept. It has five modules; the overview's diagram shows only three.

Module	What it holds	Same in every region?
1	Forms, labeling, cover letter	No, regional
2	Summaries; 2.3 is the Quality Overall Summary (QOS) of Module 3	Yes
3	Quality / CMC data, the target of this project	Yes
4	Animal study reports	Yes
5	Clinical study reports	Yes

Module 2.3 re-states Module 3's key numbers in short form. That is why "Module 2 and Module 3 disagree" is a classic error: the same number lives in two places, typed twice.

Modules 3 to 5 hold the evidence, Module 2 summarises it, Module 1 is regional

ICH M4 Common Technical Document, as filed electronically (eCTD)



Dashed border: regional, not harmonised · blue: where this project starts

The whole eCTD · 5 modules

The CTD is usually drawn as a pyramid: Modules 3, 4 and 5 are the full evidence (quality, animal studies, human studies), Module 2 condenses each into overviews and summaries, and Module 1 wraps it all in region-specific forms and labeling. Structure from ICH M4.

Inside Module 3 (the ICH M4Q layout), with the source systems each part draws on:

Section	What it proves	Typical data sources
3.2.S Drug Substance	The active ingredient: how it is made, characterized, tested (S.4), how long it stays stable (S.7)	ELN, MES/ERP batch records, LIMS, Empower
3.2.P Drug Product	The finished medicine (tablet, vial)	Same, plus formulation data

Section	What it proves	Typical data sources
3.2.P.5 Control of Drug Product	Specifications (P.5.1), test methods (P.5.2), method validation (P.5.3), batch results (P.5.4), impurity characterization (P.5.5), why the limits are justified (P.5.6)	LIMS, CoAs, Empower
3.2.P.8 Stability	Shelf life, from years of storage tests	LIMS stability module
3.2.A Appendices	Facilities (A.1), virus and contamination safety (A.2, key for biologics), new excipients (A.3)	QA documents
3.2.R Regional	Region-specific items, such as executed batch records for FDA	ERP / MES, QA

Vocabulary you will meet everywhere:

- **Batch / lot**: one production run, with an ID like 2024-07.
- **Specification**: the list of tests and pass limits for a batch, for example "Assay 95.0–105.0% of label claim".
- **CoA (Certificate of Analysis)**: one batch's test results against its specification. Contract manufacturers (**CMOs**) send these, often as scanned PDFs.
- **Assay / potency**: how much active drug is present, or how strongly it acts.
- **Impurity / degradation product**: unwanted substances from manufacture or from the drug breaking down over time.
- **Stability study**: samples stored at set temperature and humidity and tested at fixed months to set shelf life. Small molecules: long-term 25°C/60% RH, accelerated 40°C/75% RH for 6 months. Biologics: usually 5°C long-term.
- **OOS vs OOT**: out-of-specification means a result broke a limit (a formal investigation follows). Out-of-trend means it is still within limits but breaks the batch's own pattern, an early warning.
- **HPLC**: a lab instrument that separates a sample into components. Each component is a peak on a chart (a chromatogram); peak area gives the amount.
- **cGMP**: current Good Manufacturing Practice, the legal quality rules for manufacture.
- **21 CFR Part 11**: the US rule for when electronic records and signatures count as trustworthy (section 18).

- **ICH**: the body that writes the shared technical guidelines (Q1 stability, Q2 method validation, Q3 impurities, Q5/Q6 biologics, and so on).

The people involved: Regulatory Affairs owns the submission. TechOps/MSAT engineers run and scale the process. Analytical Development and QC run the tests. QA audits compliance. Medical writers turn data into narrative. Regulatory Operations (publishing) assembles the eCTD. CMOs are outside companies that make batches and send CoAs.

3. The actual problem

The hard part is not writing sentences. It is keeping thousands of numbers identical across hundreds of tables, when each number was copied by hand from a different system. One major filing draws on roughly 500 to 5,000 source documents (the overview's figure, plausible for a BLA).

Today's workflow, and where each step breaks:

#	Step	Who	Tools	What typically goes wrong
1	Plan sections, set the data-freeze date	RA project manager	Smartsheet, Teams	The freeze slips; new data lands after tables are already written
2	Pull raw data	QC scientist, MSAT lead	Empower, LIMS, Benchling, SAP, CMO portals	Exports taken on different dates; CMO results arrive as scans
3	Consolidate into Excel	Medical writer	Excel, Acrobat	Re-keying errors, unit mix-ups (µg/mg vs %w/w)
4	Write the narrative	Medical writer	Word, Veeva	Text says "all results met specification" while a table shows one that did not
5	Recalculate statistics	QC auditor	Excel	Formula drift, inconsistent rounding

#	Step	Who	Tools	What typically goes wrong
6	Check every number against source	QA auditor	Acrobat, highlighters, two monitors	Fatigue: thousands of numbers checked by eye; errors loop back to step 3
7	Publish the eCTD	Regulatory operations	Vault RIM, docuBridge	Broken links, wrong document lifecycle
8	Sign and submit	VP Regulatory, QA head	Vault, FDA gateway	A late-found error forces re-work of everything downstream

Steps 3 to 6 are what this project automates. Steps 1, 7 and 8 stay human-led, with the system feeding them.

Where the errors come from:

- **Transcription**: 0.14% typed for 0.014%.
- **Units**: the same impurity reported in %, ppm or µg/mg by different labs.
- **Version drift**: LIMS was corrected after the Word table was built.
- **Duplication**: one number lives in 3.2.S, 3.2.P and Module 2.3, typed separately each time.
- **Rounding**: results must be rounded to the specification's decimal places; people and Excel round differently.
- **Missed trends**: a slow rise in an impurity across 24 months that no one plotted.

What it costs, with the overview's numbers checked:

Overview claim	What the evidence says	Verdict
Delay costs "$1M to $3M per day"	Tufts CSDD (2024): about $800K per day in lost sales on average; medians of $1.4M (cardiovascular), $1.3M (hematology), $840K (oncology)	True for big drugs only; use ~$0.8M as the average

Overview claim	What the evidence says	Verdict
One decimal error causes a Refuse-to-File (RTF) letter	RTF is decided within 60 days of submission and is meant for applications so incomplete that review is impossible. A single wrong number more often triggers an Information Request, or questions that pile into a Complete Response Letter at the end	Overstated
An IR or RTF "halts the review clock for 30 to 90 days"	Information Requests do not stop the clock. A large response can count as a major amendment, which pushes the goal date out 3 months. An RTF is worse: you fix and resubmit, and the clock restarts	Imprecise
"Up to 12,000 human hours per filing"	No public source found	Treat as an assumption
6 to 12 months to compile a major filing	Consistent with industry descriptions	Plausible

For an IND the equivalent risk is not RTF but a **clinical hold**: FDA has 30 days to object before trials may start.

The deeper risk is credibility, not one typo. FDA expects data to meet **ALCOA+**: Attributable, Legible, Contemporaneous, Original, Accurate, plus Complete, Consistent, Enduring and Available. A reviewer who finds numbers that do not match starts doubting everything else in the file. That is why the project's core promise is **traceability**, not speed: every number must lead back to the original record.

4. Why the problem persists

The data is split across five or six systems that were never designed to talk to each other, and each one is a **validated** system that companies are afraid to touch. Here is each system, in plain words:

System (example product)	What it is	What it holds for CMC	How data gets out	Why it hurts
CDS: Waters Empower 3	Software that runs HPLC instruments and stores raw signals and peak results	Peak areas, % impurities, assay values, chromatograms	PDF/CSV reports; the Empower Toolkit programming interface, which lab-data platforms such as TetraScience use to pull results as JSON	No simple REST API, so the overview is right that it is hard, but "no hooks at all" is too strong
LIMS: Thermo SampleManager, LabWare	Tracks every sample, test, result and specification	Release results, stability study matrices, specs	SQL views, XML/CSV exports, REST on newer versions	Each site configures its own fields, so the same test has different names
ELN: Benchling	Digital lab notebook	Formulation and process development experiments	Good REST API (JSON)	Much of it is free text
ERP / MES: SAP S/4HANA, MES tools	Business and shop-floor execution systems	Batch genealogy, materials, sites, dates, batch records	SAP OData services and BAPIs	Batch records from outside manufacturers are often PDFs

System (example product)	What it is	What it holds for CMC	How data gets out	Why it hurts
DMS / RIM: Veeva Vault RIM, QualityDocs	Controlled-document and submission system	Reports, CoAs, final eCTD documents	REST API, webhooks	Knows a document's metadata and approval state, not the numbers inside it
CMO portals and email	Outside manufacturers	CoAs, batch records	Manual download	Scanned, every layout different

Five root causes:

1. **Validation lock-in.** Every system above is validated for GMP use. Connecting or changing it means re-validation paperwork, so companies leave them alone and move data by export and copy-paste.
2. **Acquisitions.** A company that bought three biotechs may run three different LIMS, each configured differently.
3. **Outsourcing.** CMOs do not give sponsors access to their systems. They send PDFs. The overview's "over 60% of CMO data arrives as scans" has no source, but the direction is right.
4. **Data-integrity rules.** FDA's cGMP rules and its data-integrity guidance expect every reported result to be attributable to an original record (ALCOA+). A script that silently copies numbers is itself a GMP system that must be validated and audited, so "just automate it" is not allowed without the audit machinery this project builds.
5. **Nobody owns the seams.** Each department owns its own system. The cross-system consistency problem belongs to the writers and QA at the very end, when changes are most expensive.

Check: the overview credits traceability to 21 CFR Part 11 alone. Part 11 governs electronic records and signatures. The "trace every number to the raw result" expectation also comes from cGMP record rules (21 CFR 211) and FDA's data-integrity guidance. Worth saying correctly in an interview.

5. Why existing tools fail

Each older tool solves one slice. None of them both **understands** the content and **proves** every number. That combined gap is the project's opening, and in 2026 several AI companies are racing for it too.

Tool family	What it does well	Where it breaks for CMC	Is the overview fair?
Document management (Veeva Vault RIM, MasterControl, Documentum)	Versioning, approval routing, e-signatures, eCTD publishing	Treats a document as a sealed box; never checks the numbers inside	Fair, but dated: Veeva announced in-product AI agents in May 2026 (see below)
Traditional OCR (Tesseract, older ABBYY)	Reads clean printed text	Tables of varying layout, multi-page tables, 1.2×10^{-3} read as "1.2x10-3", μ read as u, merged cells, tables printed over charts. Worst of all, it fails **silently** with a plausible wrong number	Fair for Tesseract. But it picks the weakest rival: modern layout-aware OCR (Azure Document Intelligence, AWS Textract, Google Document AI, open-source Docling) is much better and is a real alternative (section 12)
RPA (UiPath)	Repeats recorded clicks and copy-pastes	Breaks when a screen changes; cannot tell that "Related Substance A" and "Impurity 1" are the same compound; cannot judge	Fair. RPA can **move** data; it cannot **understand** or **verify** it

Tool family	What it does well	Where it breaks for CMC	Is the overview fair?
Chatbot LLM or basic RAG	Reads messy text, writes fluent prose	Arithmetic is not guaranteed; fixed 512-token chunks cut tables so values lose their column headers; no pointer to the source cell; can invent citations; answers change run to run and when the vendor updates the model	Fair. The last point, non-determinism, is the biggest problem for validation, and the overview barely mentions it

A note on the jargon. The overview says chatbots lack "cryptographic bounding-box provenance linking every character to the pixel coordinate". In plain words the requirement is: every reported value must link to the exact place in the source record it came from, and that link must be tamper-evident. Sections 12 and 18 show how.

What the overview leaves out: the 2026 competition. "Existing solutions" is not just Veeva and OCR anymore.

Company	What it offers	When
Celegence (CAPTIS)	Generative-AI drafting of CMC Module 3. Pilot with Kenvue claimed 80–95% alignment with previously approved text and 60% faster drafts	Announced July 2025
SyncIQ	Module 3 and Module 2.3 drafts with synchronized cross-module consistency	On the market by mid-2026
Weave Bio	AI regulatory platform across Modules 1–5, NDA workflows co-developed with Parexel	On the market by mid-2026
Peer AI	Specialized agents for preclinical, clinical and CMC documents	On the market by mid-2026

Company	What it offers	When
Veeva	Vault AI (Aug 2026), Falcon agentic platform (late 2026, early adopters), Agentic Authoring of submission documents (late 2027)	Announced May 2026

What this means for the project: "AI drafts Module 3" is no longer unique. The defensible edge is the part most drafting tools under-invest in: **deterministic verification of every number, cell-level provenance, and a validation-ready audit trail.** If you pitch this project, lead with that, not with drafting.

6. The core design principle: facts first, prose second

If you remember one thing about this project, make it this: **the language model never produces a number.** Numbers come from a store of verified facts; the model only writes the words around them. Every other component exists to fill, check, or display that fact store.

What a "fact" is. One reported value plus everything needed to trust it:

```
{
  "fact_id": "F-000482",
  "value": "0.11",
  "unit": "%",
  "analyte": "Total degradation products",
  "test": "Related substances by HPLC, method AM-102 v3",
  "batch": "2024-12",
  "condition": "25C/60%RH",
  "timepoint_months": 36,
  "limit": "NMT 0.5%",
  "source": {"system": "LIMS", "record": "ST-2024-9901/R-7731", "doc":
"CoA_2024_12.pdf", "page": 4, "bbox": [342, 110, 368, 280]},
  "checks": {"lims_matches_coa": true, "double_extraction_agrees": true,
"human_verified": false}
}
```

The pipeline, in order:

1. **Extract** candidate facts from every source, each with its source pointer.

2. **Validate** them: known unit, plausible range, and the same value from two routes (LIMS says 0.11, the scanned CoA also says 0.11). Disagreements go to a human queue.
3. **Store** them in one database of record, the fact store.
4. **Compute** with fixed, tested functions over fact IDs (mean, trend slope, shelf-life estimate). Results are new facts that record which inputs and which code version produced them.
5. **Write**: the model receives facts by ID and writes text with placeholders, for example `Total degradation products at Month {{F-000482.timepoint}} reached {{F-000482}}, within the limit of {{F-000482.limit}}.` A renderer fills in the formatted values using the specification's rounding rules.
6. **Check**: a validator scans the final text. Every digit must trace to a fact, and every claim such as "all batches met specification" is re-tested by code. Anything unmatched blocks publication.
7. **Review and sign**: a human clicks each number, sees its source, and signs. Every step is logged.

Why this is the right design: it turns a probabilistic AI problem into a normal software problem that can be unit-tested and validated. Only two steps stay probabilistic: extraction (caught by cross-checks and human review) and the wording of prose (caught by claim checks, and never the evidence itself).

Check: the overview leaves the model in charge of numbers in two places.

- Its math example has the LLM **paste the data values into the Python script** by hand (`'impurity_pct': [0.01, 0.01, ...]`). If the model mistypes one value while writing the script, Python computes the wrong answer perfectly. Fix: the script loads values from the fact store by ID; the model never types them.
- Its provenance example has the **model emit the provenance JSON** (page, bounding box). A model can invent a plausible page and box. Fix: provenance is attached by the system from the fact record, never generated by the model.

Both fixes keep the overview's architecture and simply move the numbers out of the model's hands.

7. From numbers to claims: one rule for the whole eCTD

The project covers the whole eCTD, Modules 1 to 5, with Module 3 as the first engine. Everything in the Module 3 design carries over: the fact store, validated code for every number, the model writing with placeholders, the validator, human sign-off, the hash-

chained ledger, MCP connectors, LangGraph, GEPA. What changes is the **unit of truth**, the sources, and the hardest problem.

	Module 3 engine	Whole-eCTD platform
Unit of truth	A **fact**: one number with its source	An **evidence item**: a fact, a claim, or a document reference, each linked to its source
Sources	LIMS, Empower, ELN, SAP, CMO certificates	Those, plus clinical data (EDC, SDTM/ADaM datasets, statistical outputs), nonclinical data (SEND, GLP study reports), the safety database, the labeling system, RIM and past regulator correspondence
Hardest problem	Every number matches its source	The **same** fact stays identical across five modules, the label and several regions, and every claim is backed by evidence
Who calculates	The validated statistics library	Clinical and toxicology statistics come from the sponsor's validated SAS/R programs; the platform reads and checks them, never recomputes efficacy results
Users	CMC writers, QA	Plus medical writers, biostatisticians, nonclinical scientists, labeling, regulatory operations, regulatory strategy
Data sensitivity	Trade secrets	Plus patient-level data, so privacy law now applies (section 26)
Output	Module 3 sections	A content plan, drafts for all modules, the label, cover letters, pre-publishing checks, answers to regulator questions

The rule, widened: every claim traces to evidence. Three kinds of claim, each with its own check:

1. **Numeric claims** ("the response rate was 42.1%") must match a fact taken from a validated output table. Same mechanism as section 6.
2. **Qualitative claims** ("no new safety signals were observed") must point to the evidence set they summarize. Rules check them where possible (any serious adverse event of a new type contradicts the claim); otherwise they are flagged for expert sign-off.
3. **Cross-references** ("see Section 2.7.4", "Study 301") must resolve to a real place in the dossier.

One honest boundary: the model now drafts in the most sensitive places, such as the clinical overview and the label. Benefit-risk conclusions are the sponsor's scientific judgment. The platform drafts, proves and flags; named experts own the conclusions and sign. That keeps the AI inside the "drafting a submission" use that FDA's draft AI guidance places outside its scope.

8. The evidence graph and consistency engine

This is the core differentiator. Competitors link sentences to their sources, and some reuse a value across a document set. This platform knows that "the dose in the label", "the dose in Study 301's report" and "the dose in the clinical overview" are **one fact**, stored once, shown in many places. Change it once and every place updates or gets flagged.

What the graph holds:

- **Facts**: single values with their source (a cell in an output table, a row in a dataset, a box on a scanned page).
- **Claims**: statements, each linked to the facts and evidence sets that support it.
- **Placements**: every spot in the dossier where a fact or claim appears (section, paragraph, table cell, label line, region).
- **Derivations**: which code, and which code version, produced a value from which inputs.

Consistency rules, examples:

Fact	Appears in	Rule
Dose and regimen	Protocols, CSRs, 2.5, 2.7, label section 2, cover letter	Identical everywhere
Number of patients exposed	2.7.4, integrated safety summary, label section 6	Identical, taken from the subject-level dataset (ADSL)

Fact	Appears in	Rule
Primary endpoint result	CSR section 11, 2.5, 2.7.3, label section 14	Identical to the output table, same rounding
NOAEL and safety margins	Module 4 reports, 2.4, 2.6, label sections 8.1 and 13	Consistent; margins recomputed by code
Shelf life and storage	3.2.P.8, 2.3, label section 16	Identical
Composition and excipients	3.2.P.1, label section 11	Identical
Drug substance specifications	3.2.S.4.1, 2.3.S	Identical

(Label section numbers follow the US prescribing-information format: 2 dosage, 6 adverse reactions, 8 specific populations, 11 description, 13 nonclinical toxicology, 14 clinical studies, 16 storage.)

Change propagation, the "living dossier": when a source changes (a corrected dataset, a new stability time point, a re-run table), a graph query lists every placement that depends on it. Values written as placeholders update themselves; prose around them is flagged for re-review; the change and its reach are logged. Today this impact check is done by people searching hundreds of documents.

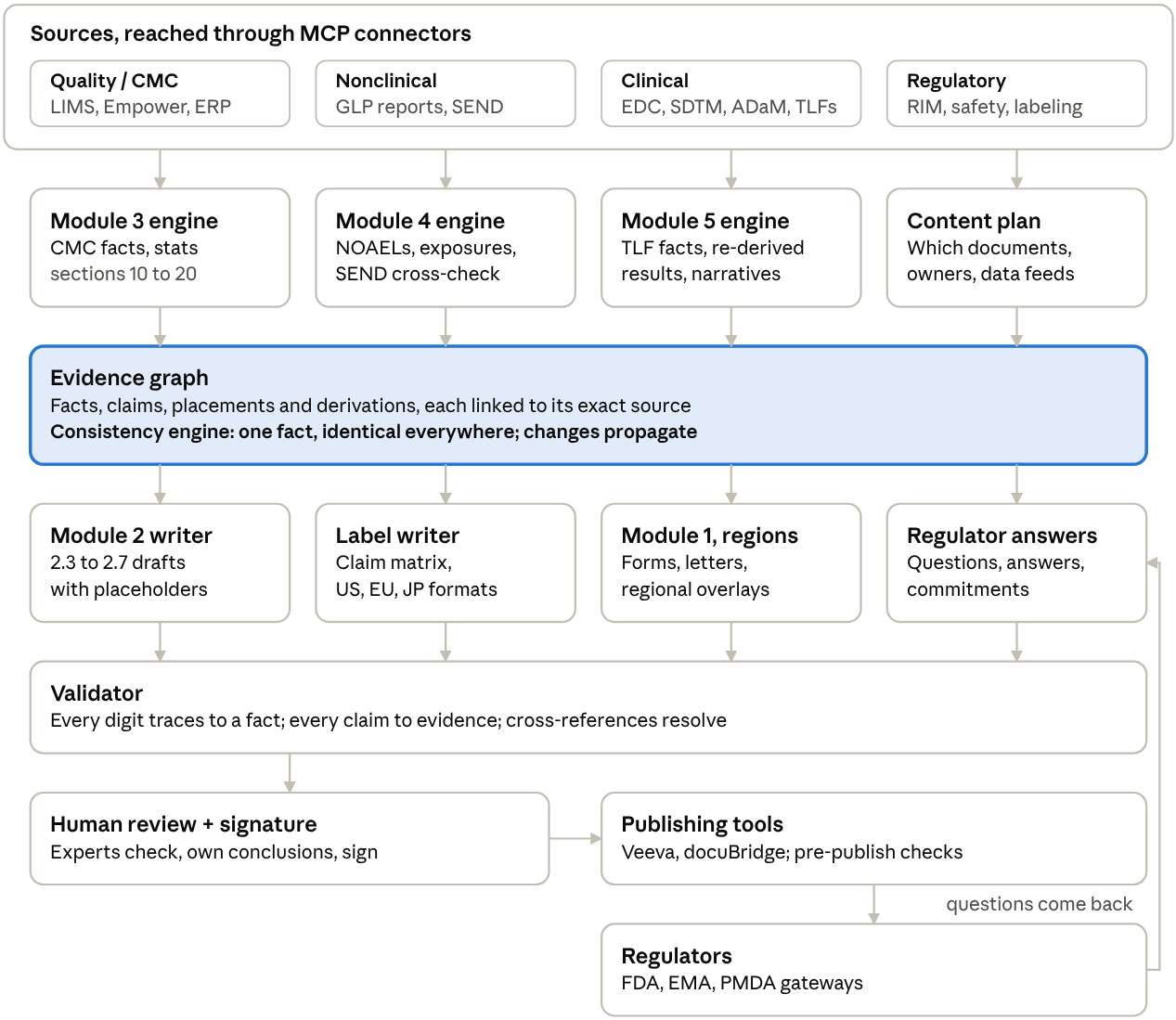
Why it is hard to copy: the value grows with every module connected. A tool that only drafts documents has nothing to propagate through.

9. The whole platform at a glance

Three module engines and a content plan feed one evidence graph; four writers turn that graph into documents; the validator, human review and existing publishing tools carry them to regulators.

Module engines feed one evidence graph; every document is written from it

Orchestrator, audit ledger and tuned prompts work as in the Module 3 engine (sections 10 to 19)



Whole-eCTD platform architecture

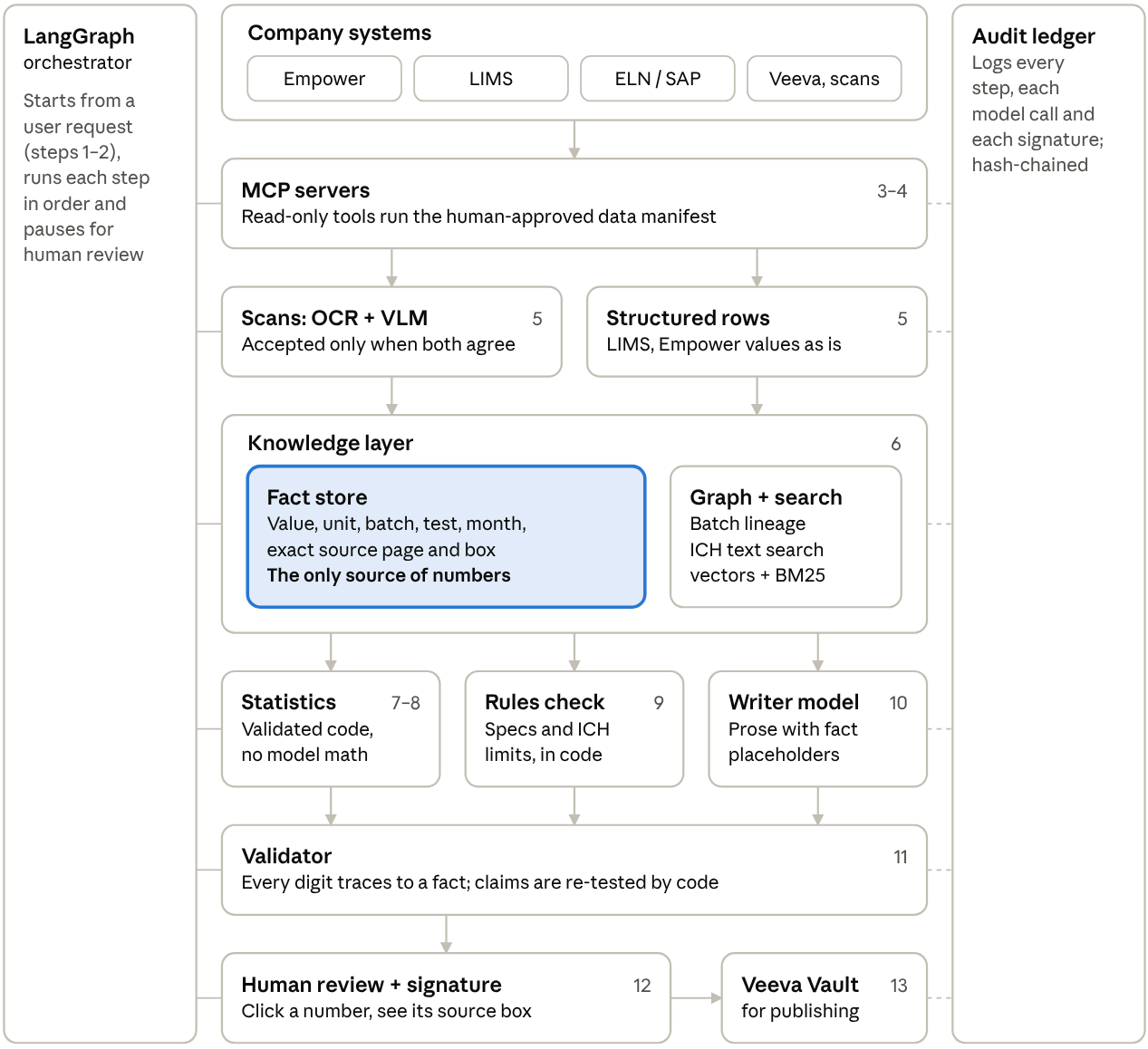
Modules 3, 4 and 5 hold evidence, so each gets an engine that turns raw sources into facts. Modules 1 and 2, the label and regulator answers are written **from** the graph, which is why they stay consistent. When regulators send questions, the answers go back through the same graph. Sections 10 to 20 now open up the first engine, Module 3, box by box; sections 21 to 26 cover the other modules.

10. The Module 3 engine at a glance

Everything is arranged around one database of verified facts. Data flows down the middle; an orchestrator on the left runs each step in order, and a ledger on the right records every one.

Every number flows through the fact store; the model only writes words

Step numbers match the walkthrough in section 20



Solid ties: steps the orchestrator runs · dashed ties: records written to the ledger

System architecture · 13 steps

Company data enters through MCP; scans are parsed and cross-checked while structured rows pass straight through; everything lands in the fact store. Statistics, rules and the writer all read from it, the validator checks the draft against it, and a human signs before Veeva publishes. Sections 11 to 19 take each box in turn.

11. Integration layer: MCP

MCP is the right plug for letting agents reach company systems, but it does not remove the hard work behind the plug. Someone still writes the code that talks to Empower, LIMS and SAP.

What MCP is. The Model Context Protocol is an open standard, created by Anthropic in November 2024 and donated on 9 December 2025 to the Agentic AI Foundation under the Linux Foundation (co-founded with Block and OpenAI). It defines how an AI application (the **client**) discovers and calls functions (**tools**), reads data (**resources**) and uses templates (**prompts**) offered by a **server**. Each tool has a name and a typed input/output schema, for example `get_stability_matrix(study_id)`. At that announcement MCP had over 10,000 active servers and 97 million monthly SDK downloads, with support in Claude, ChatGPT, Gemini, Copilot and VS Code.

Analogy: MCP is a USB-C port. Every device uses the same plug shape, but each device still needs its own driver. Here, each enterprise system needs its own MCP server (the driver).

Why MCP here:

- **Fewer integrations.** Without a standard, 4 systems × several agents × 2 models means many custom wirings. With MCP you build 4 servers, and any MCP-capable model or agent can use them.
- **Model swapping.** The same servers work with Claude or a local model, which the overview's fallback design needs.
- **Discovery and typing.** Agents see which tools exist and what arguments they take, which reduces malformed calls.

What MCP does not do (the overview overclaims "replaces custom API glue"):

- The glue moves **inside** the servers: Empower Toolkit calls, SAP OData queries, LIMS SQL. That is still most of the integration effort.
- It does nothing for data meaning: units, synonyms, site-specific field names.

- It widens the attack surface. A document returned by a tool can carry hidden instructions (prompt injection). Mitigations: read-only tools, least-privilege service accounts, per-user OAuth scopes, an allow-list of tools per agent, and logging every call.
- It is not deterministic by itself. In a GMP setting the data pull for a section should be a **reviewed data manifest** (these studies, these batches, these documents) that tools execute exactly. The model may propose the manifest; a human approves it.

Alternatives, and why not (or why also):

Option	Strength	Weakness	Verdict
Custom REST adapter per system	Full control	Each agent and model needs its own wiring; no discovery	MCP servers are exactly these adapters with a standard front
RPA screen-scraping	Works when there is no API	Fragile, hard to validate	Last resort for UI-only systems
ETL / iPaaS (MuleSoft, Informatica) into a data lake	Enterprise-grade bulk sync	Built to move data, not for on-demand agent calls	Good **complement**: ETL fills the fact store in bulk, MCP serves agents on demand
Lab-data platform (TetraScience)	Already connects Empower and LIMS, outputs JSON	Cost, vendor lock-in	Strong "buy" option for the Empower connector
One vendor's function calling	Simple	Locks you to one model API	MCP is the vendor-neutral layer over it

Check: the Milestone 1 exit criterion, "retrieval in under 500 ms", measures the wrong thing. Enterprise LIMS and CDS queries often take seconds, and speed is not the risk. A better criterion: every record of a reference study is retrieved, with record counts and checksums matching a manual export.

12. Document parsing: the vision-language pipeline

Use a vision-language model (VLM) to **understand** a scanned table, but never trust it alone for the digits or the coordinates. Pair it with a layout-aware OCR engine and accept a value only when both agree.

Rule zero: prefer data over pixels. If a value exists in LIMS or Empower, take it from there. Parse PDFs only for what exists only as PDF: CMO Certificates of Analysis, legacy reports, scanned batch records.

Check: the overview has the VLM read peak areas off chromatogram **charts**. That is the wrong source. Empower already stores the integrated peak areas as numbers, and a CMO's chromatogram report prints them in a results table under the curve. Extract the table. A VLM looking at the curve can help as a sanity check ("the table lists 6 peaks, the chart shows 6"), never as the source of a reported value.

Two kinds of PDF need different handling:

- **Born-digital** (exported from software): the text and exact coordinates are already inside the file. Libraries such as pdfplumber or PyMuPDF read them deterministically. No AI needed for the digits.
- **Scanned** (a photo of paper): only pixels. This is where OCR and VLMs come in.

The recommended pipeline for scans:

1. **Classify** the document (CoA, batch record, stability report) and check scan quality (resolution, skew).
2. **Layout + OCR**: find tables, headers and footnotes, and read every word with its box on the page. Tools: Docling (open source, runs locally), Azure Document Intelligence, AWS Textract, Google Document AI.
3. **VLM interpretation**: the model reads the page image plus the OCR words and outputs structured JSON. It decides which column is Month 12, that a footnote says results are in µg/mg, that "Related Substance A" is the known impurity 1. Crucially, it points to **OCR word IDs**, not free-hand coordinates.
4. **Cross-check**: each value the VLM outputs must equal the OCR word at the location it points to. VLM says 0.11 but the OCR word there reads 0.14? Flag it.
5. **Route by confidence**: disagreement or low confidence goes to a human queue.
6. **Normalize**: units (a units library such as pint), scientific notation, and special results such as "<0.05%" (below reporting threshold), "ND" (not detected), "NMT" (not more than), "Complies".

Why a VLM at all, and not just OCR? OCR returns characters. It does not know that a table continues on the next page, that a merged header spans three columns, or that a footnote changes the unit. VLMs handle that layout meaning well.

Why not a VLM alone? VLMs can misread a digit in a poor scan, and they are imprecise at pixel coordinates. Asking a VLM for bounding boxes (the overview's `[ymin, xmin, ymax, xmax]` format) gives approximate boxes. For audit-grade provenance, the box must come from the OCR layer and the VLM must agree with it.

Tool	Type	Good at	Weak at
pdfplumber, PyMuPDF	Digital-PDF readers	Exact text and coordinates, free, deterministic	Scans
Tesseract	Classic OCR	Free, local	Tables, symbols
Docling	Open-source layout + table model	Table structure, runs inside a locked-down VPC	Very poor scans
Azure DI, Textract, Document AI	Managed layout OCR	Strong tables with coordinates	Cloud dependency
Claude, Gemini, GPT (vision); Qwen-VL (local)	VLMs	Layout meaning, footnotes, synonyms	Exact digits in bad scans, pixel boxes

On chemical structures: the overview lists "non-standard chemical structures" as a parsing target. Structure drawings in Module 3 are produced by the sponsor and already exist as structure files in the ELN or compound registry, so there is rarely a need to read them from images. If needed, specialized structure-recognition models exist; general VLMs are unreliable at converting a drawing into an exact molecule.

Check: ">99.5% extraction accuracy on 2,000 documents" needs a definition. At field level, a CoA with 200 values would still average one error per document, so humans must verify anyway. The metric that matters is **zero undetected errors**: every wrong extraction was flagged by the cross-check or caught in review. Measure how well the flagging catches errors (its recall), not raw accuracy.

13. Knowledge layer: Neo4j + Pinecone + BM25

The system asks three different kinds of questions, and each needs a different kind of index. That is the real justification for "hybrid". The overview's specific products, though, clash with its own security design.

Question type	Example	Best index	Why
Relational, multi-hop	"All batches made at Site B with buffer lot X that were out of trend for dissolution"	Graph (or SQL joins)	Follows links: batch → site, batch → material lot, result → test
Meaning	"What does ICH say about qualifying a degradation product above threshold?"	Dense vectors (embeddings)	Finds passages with the same meaning in different words
Exact term	"Q3B", "AM-102", "ST-2024-9901"	Keyword search (BM25)	Embeddings blur codes and IDs; keywords match them exactly

The graph, concretely. Things become nodes, relationships become edges:

```
(Product)-[:HAS_BATCH]->(Batch)-[:MADE_AT]->(Site)
(Batch)-[:USED_LOT]->(MaterialLot)
(Result)-[:FOR_BATCH]->(Batch)
(Result)-[:OF_TEST]->(Test)-[:USES_METHOD]->(Method {version})
(Result)-[:AT]->(Condition {temp, rh}), (Result)-[:AT_MONTH]->(Timepoint)
(Result)-[:EXTRACTED_FROM]->(SourceLocation {doc, page, bbox})
(Section {id:'3.2.P.5.4'})-[:CITES]->(Result)
(Impurity)-[:SYNONYM_OF]->(Impurity)
```

The relational question above becomes one Cypher query:

```
MATCH (b:Batch)-[:MADE_AT]->(:Site {name:'Site B'}),
      (b)-[:USED_LOT]->(:MaterialLot {id:'BUF-X'}),
      (r:Result {oot:true})-[:FOR_BATCH]->(b),
      (r)-[:OF_TEST]->(:Test {name:'Dissolution'})
RETURN b.id, r.value, r.month
```

Where a graph earns its place: batch genealogy (which drug-substance batch went into which drug-product batch), synonym resolution, and impact analysis ("method AM-102 changed to version 4: which dossier sections cite results from version 3?"). That last one is what powers the post-approval future scope.

Honest counterpoint. The numbers themselves are perfectly tabular: batch, test, condition, month, value. A relational table handles them more simply, with exact types, constraints and transactions. PostgreSQL, already in the stack for the audit ledger, can also do vector search (pgvector) and full-text search. A strong version 1 keeps the **fact store in Postgres** and adds a graph database only when multi-hop lineage queries become central.

Check: "Graph RAG" means two different things. Microsoft's GraphRAG (2024) has an LLM extract entities and relationships from free text. This project's graph should be built from **structured source data** by deterministic code. Never let an LLM invent edges between GMP facts.

How retrieval should work: a query runs through BM25 and vector search in parallel; results are merged (reciprocal rank fusion), re-scored by a reranker model, and filtered by metadata such as guideline, region and version. Versions matter: the ICH Q1 revision reached Step 2 (public draft) in April 2025 and will supersede Q1A–Q1F and Q5C, so the index must know which text applies to a given filing.

The real fix for "chunking destroys tables" is table-aware chunking: keep a table whole, or chunk by row with its column headers repeated. Guidelines are chunked by their own numbered sections. Long context helps, but this is the actual cure.

Check: three sovereignty contradictions. The overview promises a single-tenant, isolated AWS GovCloud deployment, then picks:

- **Pinecone Serverless**, a multi-tenant service run in Pinecone's own cloud. Pinecone's BYOC option (generally available, Enterprise plan) puts the data plane in your AWS, GCP or Azure account, but its docs do not list GovCloud.
- **Neo4j Aura**, Neo4j's managed service on commercial clouds. For isolation you would self-host Neo4j Enterprise inside the VPC.
- **OpenAI text-embedding-3-large**, an external API. Every chunk of proprietary text would be sent to a third vendor to be embedded.

Need	Overview choice	Recommended inside a locked-down VPC
Fact store (the numbers)	Implicitly Neo4j	PostgreSQL tables
Relationships, lineage	Neo4j Aura	Self-hosted Neo4j Enterprise, or Postgres first
Meaning search	Pinecone Serverless	pgvector or OpenSearch (millions of chunks is small for either)
Keyword search	BM25 via Pinecone sparse vectors	OpenSearch BM25 or Postgres full-text
Embeddings	OpenAI text-embedding-3-large	An open embedding model on the project's own GPUs

14. Orchestration: LangGraph and the agents

LangGraph is a good choice, but for the right reason: it lets you build a **mostly fixed, auditable state machine** with a few AI steps inside, rather than a crowd of free-roaming agents. In a GMP setting, predictable beats clever.

What LangGraph is. A library from the LangChain team for building AI workflows as graphs:

- **Nodes** are steps (a Python function or a model call).
- **Edges** connect them; conditional edges choose the next step from the current state.
- **State** is one typed object that every node reads and updates.
- A **checkpointer** saves the state after every step, so a crashed run resumes where it stopped.
- **Interrupts** pause the graph for a human to review, edit or approve, then resume.

LangGraph 1.0 became generally available on 22 October 2025, highlighting durable state, built-in persistence and human-in-the-loop, with Uber, LinkedIn and Klarna named as production users. **Check:** the overview's "LangGraph v0.3+" is outdated; target 1.x.

Workflow vs agent. A workflow follows paths you wrote in code; an agent lets the model decide the next step. This pipeline is mostly a workflow: ingest → parse → validate → compute → retrieve rules → draft → check → human review → sign. Model freedom belongs

only inside bounded steps, such as interpreting an odd CoA layout or phrasing a paragraph. The "supervisor" should therefore be mostly routing code with a model consulted for exceptions, not a model improvising the whole run.

The agents, judged:

Agent	Job	Should be	Comment
Ingestion	Fetch data via MCP	Deterministic	Executes the human-approved data manifest (section 11)
VLM Parser	Turn scans into facts	Model + deterministic cross-check	Section 12
Mathematical Verification	Statistics and table-math checks	Deterministic code	The model may pick which validated function to call, never compute or type values
Regulatory Compliance	Compare results with specs and ICH thresholds; retrieve and explain guidance; check section completeness against M4Q	Rules for comparisons, model for retrieval and explanation	"0.11% vs limit 0.2%" is a code comparison, not a model judgment
Compliance Audit	Record every step	Not an agent	Logging must be automatic at framework level (callbacks on every node and tool call). An agent that decides what to log is a compliance hole

Agent	Job	Should be	Comment
Redaction (short overview only)	"PII / HIPAA filter"	Rules + model	**Check:** HIPAA protects patients' health information, which CMC data almost never contains. What CMC files do contain is employee names and signatures in batch records and CMO trade secrets. Reframe as confidentiality redaction
Missing: Writer	Draft prose with fact placeholders	Model	The overview folds this into the supervisor; it deserves its own node
Missing: Consistency Checker	Same value identical in 3.2.S, 3.2.P and Module 2.3; text claims match tables	Deterministic	The single biggest error source in section 3, yet no agent owns it

Why LangGraph and not the alternatives:

Option	What it is	Fit here
LangGraph	Graph state machine with checkpoints and interrupts	Explicit control flow, resumable, human-in-the-loop built in. Downside: a fast-moving ecosystem with extra abstraction
CrewAI	Role-based agent teams	Built around autonomous collaboration; control flow is less explicit, so harder to validate

Option	What it is	Fit here
AutoGen-style chat agents	Agents converse to solve a task	Conversation paths vary run to run; poor for validation
Temporal, Airflow, Prefect	Durable workflow engines	Excellent retries, durability and history. A serious option for the **outer** pipeline, with LangGraph inside individual steps
Plain Python + a task queue	Your own code	Maximum control, but you rebuild checkpointing and human pauses yourself
Vendor agent SDKs (Claude Agent SDK, OpenAI Agents SDK)	One vendor's agent toolkit	Tied to one model provider; fine for sub-tasks

Two design corrections:

- **Checkpoints in Postgres, not Redis.** The overview checkpoints to Redis, an in-memory store. For a regulated audit trail, checkpoint to Postgres (LangGraph supports it), and hash each checkpoint into the ledger.
- **Retry only transient failures.** "Automated error recovery loops" are fine for timeouts. They are dangerous for failed verifications: a model re-trying until the checker passes learns to satisfy the checker, not the truth. A failed number check goes to a human.

15. Models: Claude Opus 4.6 and DeepSeek-R1

A strong frontier model for interpretation and writing plus a local open model for sovereignty is a sound pattern. The specific pairing in the overview has three problems: the cloud model cannot run inside the promised GovCloud boundary, R1 cannot see images, and both picks are already a generation old.

Why a frontier model like Claude for the main role: long context (read a whole study report at once), strong reasoning over scientific text, vision for scanned pages, reliable structured output (JSON, tool calls), and high-quality technical writing.

What Claude Opus 4.6 actually is. Released 5 February 2026 with a 1M-token context window, in beta at launch. Anthropic reported 76% on the MRCR v2 long-context retrieval benchmark versus 18.5% for its predecessor, and large gains on life-science benchmarks

(self-reported). It is still active, with retirement not before 5 February 2027.

It is no longer the newest. As of September 2026 Anthropic's current models are Claude Opus 5.5 and Sonnet 5.5 (1M context; $4 / $20 and $2 / $10 per million input / output tokens) and Claude Fable 5.1 for the hardest reasoning. The lesson is not "switch to 5.5" but **never hard-wire a model**: pin an exact model ID per validated release, log it on every call, and re-validate before changing it. Anthropic gives at least 60 days' notice before retiring a model.

Check: the GovCloud contradiction. AWS's FedRAMP list for Bedrock models shows Claude 4.5 Opus authorized in GovCloud, but not Opus 4.6. So "Claude Opus 4.6 (Cloud)" inside a GovCloud single-tenant VPC would mean calling a public API outside the boundary, which breaks the overview's own sovereignty promise.

What DeepSeek-R1 actually is. An open-weights reasoning model released January 2025 under the MIT license: 671B total parameters (mixture-of-experts, about 37B active per token), **text only**. Three problems:

- **It cannot see.** The challenges table says the local model does "preliminary data parsing". Parsing scans needs vision; R1 has none.
- **It is heavy.** About 700 GB of weights at FP8 is more than one 8×H100 node holds (8 × 80 GB = 640 GB), so full R1 needs multiple nodes. The small "distilled" R1 versions fit easily but are much weaker.
- **It is outdated and politically awkward.** DeepSeek V4 shipped 24 April 2026 (1.6T parameters, 1M context). At least 15 US states banned the DeepSeek app on government devices in 2025. Open weights run locally send nothing to China, but many pharma security reviews flag Chinese-origin models, and a GovCloud pitch makes that worse.

Local alternatives: OpenAI's gpt-oss-120b (open weights, Apache 2.0, runs on a single 80 GB GPU), Llama and Mistral models, IBM Granite. Qwen is strong, including vision models, but is also Chinese-origin.

Check: long context does not replace retrieval. 1M tokens is roughly 750,000 words, on the order of 1,500 to 3,000 pages of dense text, and page **images** cost more tokens than text. The overview's 50,000-page ingestion cannot fit. Use long context to read one whole report or one section's evidence at once; use the fact store and retrieval for everything else. Cost also scales: at $4 per million input tokens, one full 1M-token call costs about $4 in input alone before caching.

Check: "fallback" is risky in a validated system. Switching silently to a different model when the primary is down produces different outputs. Each model configuration must be validated for its job; a switch is a controlled, logged event.

Recommended model roles:

Role	Overview pick	Recommendation
Numbers, statistics, comparisons	(LLM writes scripts)	No model: validated code (section 16)
Page interpretation (scans)	VLM, unspecified	A vision model cross-checked against OCR; a local vision model where data cannot leave
Classification, routing, extraction QA	DeepSeek-R1	A small fast model (cloud) or a local open model
Scientific drafting	Claude Opus 4.6	Current frontier model available **inside** the deployment boundary, version-pinned
Sovereign or offline mode	DeepSeek-R1 on H100	A validated local open-weights model sized to the hardware

16. Math sandbox: code, not model arithmetic

Moving math out of the model and into code is the overview's best idea. But the stronger version is a **pre-validated statistics library** the model can only call, not a sandbox where the model writes fresh code for every request.

Why code at all. A model predicts the next token. It is usually right at arithmetic, and "usually" cannot be validated. Code gives the same answer every time, can be unit-tested, and can be re-run by an inspector years later.

Check: "PyChain WASM" does not exist. No product or library by that name turns up. The overview most likely means "Python running inside a WebAssembly sandbox inside Docker". Real options:

Sandbox	What it is	Trade-off
Pyodide	CPython compiled to WebAssembly; runs NumPy, pandas, SciPy	Strong isolation, slower, some packages missing
gVisor, nsjail	Kernel-level isolation for ordinary containers	Fast, needs careful configuration
Firecracker microVMs (used by services such as E2B)	Tiny virtual machines per run	Strongest isolation, more setup
Plain Docker, no network, read-only disk, CPU/memory limits	The common baseline	Weakest of the four, often enough inside a private VPC

The better design for GMP work:

1. Write a **statistics library** once: mean, standard deviation, %RSD, ICH Q1E regression and poolability, shelf-life estimation, out-of-trend tests, specification comparisons, regulatory rounding.
2. Unit-test it against reference results and validate it once (risk-based, GAMP 5 style).
3. Expose each function as a tool: `shelf_life(fact_ids, spec_limit)`. The model chooses the function and the fact IDs; it never writes the math or types the values.
4. Keep the free-code sandbox only for exploratory, clearly labelled non-GMP analysis.

The statistics that actually apply (explained simply):

- **Shelf life (ICH Q1E):** fit a straight line of the attribute against time. For an impurity that rises, shelf life is where the **upper one-sided 95% confidence bound** of that line crosses the specification limit (for assay that falls, the lower bound). Batches may be pooled only if their slopes and intercepts do not differ, tested at a significance level of 0.25. Extrapolation beyond the real data is allowed only within limits (roughly up to twice the data period, and no more than 12 months beyond it).
- **Out of trend (OOT):** a result still inside specification but unusual for this product. Standard methods compare the new result with a **prediction interval** built from earlier data or historical batches (regression control chart, by-time-point method, slope control chart).

- **Awkward values:** "<0.05%" (below the quantitation limit) cannot simply be averaged. The library needs a written policy for it.
- **Rounding:** results are rounded to the number of decimals in the specification, rounding a trailing 5 up. **Python's built-in `round()` does not do this**: `round(0.125, 2)` gives 0.12 and `round(2.675, 2)` gives 2.67 (half-to-even rounding plus binary floating point). Use `Decimal` with `ROUND_HALF_UP`, which gives 0.13. A classic silent bug.

Check: the overview's example script has four flaws. I ran it on its own data:

1. **Values are typed into the script by the model** (section 6).
2. **Its OOT test is nearly blind.** It fits the line through all points, including the Month-36 value it is testing, then flags OOT only if that value is 0.02 above the line. Because the tested point drags the line toward itself, the test first flags OOT only when the Month-36 value reaches **0.164%**. A proper 95% prediction interval from the earlier points flags anything above about **0.124%**. The fixed 0.02 margin has no statistical basis.
3. **"Mean impurity" across time points is meaningless**: it averages a trend. Useful are the slope, or the mean across batches at one time point.
4. **Formatting to 4 decimals** ignores the specification's reporting precision.

Check: "100% mathematical accuracy" is true only of arithmetic. Code removes arithmetic errors; it does not remove wrong inputs, the wrong statistical method or the wrong rounding rule. Likewise "zero calculation errors across 500 matrices" is trivially true for deterministic code. The meaningful test compares outputs with independently validated reference results (for example from JMP or SAS) on hard edge cases: missing time points, "<LOQ" values, rounding ties, a single batch that cannot be pooled.

17. Human review workspace

The review screen is where the project earns trust: click any number in the draft and see the exact source cell, highlighted. Its hardest problem is not the code but **automation bias**: people start rubber-stamping green checkmarks.

What the screen does:

- **Left pane:** the draft section in a rich-text editor. Every number is a non-editable "fact chip" carrying its fact ID; a writer cannot type a raw number.
- **Right pane:** the source PDF, rendered with a box drawn over the exact cell when a chip is clicked.

- **Fact panel:** value, source system, extraction route, which checks passed, which calculations used it, and its review state (unverified, verified, rejected with reason).
- **Live progress** of long runs over WebSockets.

Why a human must sign. The sponsor is legally responsible for what it submits. Under Part 11 an accountable person signs electronically, and the signature shows their name, the date and time, and its meaning, such as "Approved" (section 18). An AI cannot hold that responsibility. FDA's January 2025 **draft** guidance on AI for regulatory decision-making asks for a risk-based credibility assessment of any AI model for its specific context of use; strong human review lowers the model's influence on the final decision and so lowers the evidence burden. The FDA–EMA Good AI Practice principles (14 January 2026, voluntary) stress the same things: human-centric design, a clear context of use, data governance and lifecycle management.

Designing against rubber-stamping:

- **Risk-based review.** A value pulled by API from validated LIMS, with a matching checksum, needs a lighter check than a value read from a blurry scan. Machine-read values with any disagreement always need an explicit human check. This is what makes "100% human verification" affordable.
- **Show doubt loudly.** Disagreements, low confidence and "<LOQ" handling sit at the top, not buried.
- **Challenge testing.** In a test mode, seed known errors and measure whether reviewers catch them.
- **Attest per table** for high-risk sections, not one click for the whole document.
- **Re-check on every edit.** When a writer changes text, the claim and number validator runs again.

Technology choices:

Piece	Overview pick	Why it fits	Alternatives
UI framework	React 19 + TypeScript	Largest ecosystem for complex apps; types catch bugs	Angular (common in enterprise pharma), Vue, Svelte: no strong reason to switch

Piece	Overview pick	Why it fits	Alternatives
Editor	ProseMirror	A toolkit for **structured** documents with custom node types, exactly what fact chips need (Tiptap is built on it)	Lexical, Slate; Quill is too loose for structured content
PDF viewer	(unnamed)	PDF.js renders PDFs in the browser and allows overlays	Commercial SDKs such as Apryse

Two practical traps:

- **Coordinate mapping.** OCR boxes are in image pixels; PDF.js draws in PDF points, and pages may be rotated or scaled. A wrong conversion puts the red box on the wrong cell, which destroys trust. Test it on rotated and odd-sized scans.
- **Writers live in Word.** Adoption depends on meeting them there: export to .docx with the fact links preserved as bookmarks or fields, or build a Word add-in. Veeva's announced Agentic Authoring integrates with Word for this reason.

Section 34 shows what this workspace looks like to the writer and reviewer, and the design choices behind it.

18. Audit ledger and 21 CFR Part 11

Part 11 requires secure, computer-generated, time-stamped audit trails and properly linked e-signatures. It does **not** require blockchains or hash chains. A hash chain adds tamper-**evidence**, a good extra, and the overview's append-only PostgreSQL with SHA-256 chaining is a sensible, cheap design once three holes are closed.

What 21 CFR Part 11 asks for, section by section:

Section	Requirement in plain words	How this project meets it
11.10(a)	The system is validated	GAMP 5-style validation (section 31)
11.10(d)	Only authorized people get access	SSO + role-based permissions (section 19)

Section	Requirement in plain words	How this project meets it
11.10(e)	Secure, computer-generated, time-stamped audit trails of who created, changed or deleted what; old values stay visible	The ledger
11.10(k)	Controlled system documentation	Versioned SOPs, specs, change control
11.50	A signature shows the printed name, date and time, and meaning ("Approved")	Signature records in the ledger and on the rendered PDF
11.70	Signatures linked to their records so they cannot be cut, copied or moved to falsify another record	The signature record includes the SHA-256 hash of the exact signed document
11.100 / 11.200 / 11.300	Signatures unique to one person; non-biometric ones need two components (ID + password); password controls	Re-authentication at signing, through the company's identity provider

How a hash chain works. Each entry stores a fingerprint (hash) of itself **plus the previous entry's fingerprint**:

```
h1 = SHA256(h0 + entry1)
h2 = SHA256(h1 + entry2)
h3 = SHA256(h2 + entry3)
```

Change anything in entry 1 and h1 changes, which breaks h2, h3 and every link after it. A verifier recomputes the chain from the start and finds the first break.

Three holes and their fixes:

1. **An admin can rewrite the whole chain** by recomputing every hash. Fix: periodically **anchor** the latest hash outside the database, for example in S3 with Object Lock in compliance mode (write-once storage nobody can delete, even an admin) or with an RFC 3161 timestamping service.
2. **"Append-only" must be enforced, not promised.** Fix: the application's database role may INSERT only; no UPDATE or DELETE grants; a trigger rejects updates anyway.

3. **Anyone with insert rights could append forged entries.** Fix: sign each entry (or each batch of entries) with a key held in AWS KMS / an HSM that only the ledger service can use.

What to log: every user action (who, what, when, why); every model call (exact model ID, full prompt, parameters, output); every tool call; every data pull (record IDs + checksums); the version (git commit) of the statistics library; every workflow checkpoint; every review decision; every signature with its document hash. Large payloads such as 1M-token prompts go to write-once object storage; the ledger stores their hashes.

Alternatives, and why not:

Option	Status	Verdict
Amazon QLDB (a managed ledger with built-in hash chains)	**Discontinued: end of support 31 July 2025**; AWS recommended Aurora PostgreSQL, which has audit logging but no built-in cryptographic verification	Would have been the obvious pick; a reminder that owning the design matters
Blockchain (Hyperledger and similar)	Available	Overkill: one organization, no need for distributed consensus
immudb	Open-source immutable database	Credible alternative
Azure SQL Ledger	Available on Azure	Good if the company runs on Azure
PostgreSQL + hash chain + WORM anchoring + signing key	The overview's choice plus fixes	Recommended

The European side, which the overview omits. EU GMP Annex 11 is Europe's counterpart to Part 11, and a revision was published for consultation in July 2025 together with a new **Annex 22 on AI**. Draft Annex 22 says generative AI and LLMs "should not be used in critical GMP applications"; for non-critical uses they are acceptable when qualified people confirm the output. It was still a draft in August 2026, with final text targeted for

late 2026. Drafting a submission with a human verifying every number fits the non-critical, human-confirmed pattern. The overview's future idea of an LLM adjusting bioreactors (section 32) collides with it head-on.

19. Security, deployment and scale

One isolated environment per customer is exactly right for pharma, which guards formulas and processes as crown jewels. AWS GovCloud is the wrong default: it is built for US government workloads, lags on AI model availability, and would block the European and Japanese expansion the overview plans.

The gateway layer:

Piece	What it does	Why this choice	Notes
FastAPI	Python web framework serving the app's API	The AI stack is Python (LangGraph, NumPy), so one language end to end; automatic API docs; request validation with Pydantic	It is an app framework, not an "API gateway". A load balancer with a web application firewall (or Kong/Envoy) still sits in front
OAuth2 / OIDC, SAML 2.0 via Okta or Microsoft Entra ID (formerly Azure AD)	Single sign-on with the company's own identity system	Pharma will not accept separate passwords; roles come from their directory	Roles: writer, QA reviewer, approver, admin. Re-authenticate at signing (Part 11.200)
Redis rate limiter, circuit breaker	Stops the app from flooding LIMS or Empower; trips when a dependency fails	Legacy lab systems fall over under bursts	Reasonable

Piece	What it does	Why this choice	Notes
TLS 1.3, FIPS 140-3 key management	Encryption in transit; US-government-validated crypto modules (AWS KMS HSMs)	Required for government work; harmless otherwise	Fine

Check: GovCloud. AWS GovCloud (US) exists for regulated US government data (FedRAMP High, ITAR, defense). For a pharma customer it brings a smaller service catalog, US-persons-only administration, and slower model availability: Claude Opus 4.6, the overview's own primary model, is not on AWS's GovCloud Bedrock list (section 15). Pharma companies normally run GMP systems in commercial cloud regions under a documented shared-responsibility model. A European customer will want an EU region for GDPR. Recommended: a commercial region per customer (US or EU); GovCloud only as an option for government-funded programs.

Check: "FedRAMP High, HIPAA, 21 CFR Part 11 compliant" infrastructure. No infrastructure is Part 11 compliant by itself. Part 11 compliance is a property of the whole system plus the customer's procedures, and the customer validates it. HIPAA is largely irrelevant to CMC data (section 14).

Check: Nitro Enclaves. AWS Nitro Enclaves are sealed-off environments with no storage, no interactive access and no outside network, designed for CPU work such as handling keys. They are not a place to host GPU model inference. Per-customer VPC isolation is already the main control.

The GPU layer. vLLM is an open-source inference server that makes one GPU serve many requests at once (paged attention memory, continuous batching) behind an OpenAI-compatible API; SGLang and TensorRT-LLM are alternatives. Size GPUs to the model: gpt-oss-120b fits one 80 GB H100; full DeepSeek-R1 or V4 needs multiple nodes (section 15). Newer H200 and Blackwell GPUs carry more memory per card.

Scale: asynchronous queues. Parsing is batch work, so it runs through a job queue with worker pools (the overview: Celery + Redis Streams; alternatives: AWS SQS with ECS or Batch, Temporal, Ray for GPU batches). Three rules:

- **Idempotent jobs**: key every document by the hash of its file, so reprocessing yields the same fact IDs instead of duplicates.

- **Batch pages for the GPU** (the overview's "dynamic VLM batching") to keep utilization high.
- **Back-pressure**: queue depth, not user clicks, drives load on the GPUs and enterprise systems.

For scale intuition, with illustrative numbers: 50,000 pages at about 5 seconds each across 20 parallel workers is about 3.5 hours.

Check: tenancy wording. "Strict memory enclave isolation across client tenants" contradicts "single-tenant". If each customer has its own VPC, no tenants share memory. If you ever go multi-tenant SaaS, you need a tenant ID on every record, per-tenant encryption keys, separate vector namespaces, and no shared caches, including model prompt caches.

20. One request end to end: the impurity walkthrough

The overview's example run is a good skeleton, and its sample output sentence contains four pharmaceutical science errors. That is itself the best argument for the project: even a careful expert document slipped, and a rules-aware checker would have caught every one.

The run, step by step (the step numbers match the badges on the section 10 diagram):

#	Step	What happens	Fix or note
1	Request	User picks product, section, batches 2024-01 to 2024-12 and a data-freeze date	Add: the system proposes a **data manifest** and a human approves it
2	Start	Gateway checks the login token and role, opens a run ID, writes the first ledger entry	Roles are company-defined; Part 11 does not define "permissions"
3–4	Fetch	Ingestion calls MCP tools: `get_documents` (Veeva), `get_stability_matrix` (LIMS), `get_formulation_logs` (ELN)	Executes the manifest exactly
5	Parse	Scans go through OCR + VLM; LIMS rows become facts directly	Boxes come from the OCR layer (section 12)

#	Step	What happens	Fix or note
6	Store	Facts and relationships are written to the fact store and graph	Reconcile LIMS vs CoA values first
7–8	Compute	Validated functions compute trend, OOT and shelf life	Loads values by fact ID; the model types none (section 16)
9	Rules	Regulatory agent retrieves ICH text and compares results with limits	Comparisons in code; guideline chosen by modality and dose (below)
10	Draft	Model writes the narrative with fact placeholders	Provenance attached by the system, not the model
11	Check	Number and claim validator, cross-section consistency check	Missing in the overview as a named step
12	Review and sign	Reviewer clicks each number, sees the source box, signs; document hash goes to the ledger	Risk-based review depth (section 17)
13	Export	Signed section is pushed to Veeva Vault RIM	Veeva's own publishing and approval then take over

Check: the overview's sample sentence. It reads: *"The total degradation products at Month 36 under accelerated conditions (40C/75% RH) reached 0.11%, remaining well within the ICH Q3B qualified threshold of 0.20%."*

1. **Accelerated studies do not reach Month 36.** Under ICH Q1A(R2), accelerated storage (40°C/75% RH) runs 6 months, for example at months 0, 3 and 6. Month 36 belongs to long-term storage (25°C/60% RH).
2. **Total vs individual.** ICH Q3B thresholds apply to **each individual** degradation product. Total degradation products are controlled by the product's own specification limit. Comparing a total with an individual threshold mixes two different things.
3. **The threshold depends on dose.** Q3B's qualification threshold is 1.0% below 10 mg/day, 0.5% for 10–100 mg, 0.2% for 100 mg–2 g, and 0.15% above 2 g (each "or a fixed daily intake, whichever is lower"). The 0.20% figure is only right for a 100 mg to 2 g daily dose, so the system must know the dose. The correct term is "qualification threshold".
4. **Q3B does not apply to biologics.** Q3B explicitly excludes biological and biotechnological products, yet the overview frames this as a BLA, and its stability grid (Month 1, frozen storage) looks like a biologic's. For a biologic the rules come from ICH

Q6B and Q5C.

Two data-sanity catches the checker should also make:

- **Impossible dates.** Batch 2024-12 was made in December 2024, so Month-36 data cannot exist before December 2027. A simple rule (manufacture date + time point ⩽ test date ⩽ today) catches this.
- **Wrong source type.** Month-36 stability results normally come from LIMS or a stability report, not a release Certificate of Analysis.

Check: "historical limits from previous filings for the same molecule class" (step 9) is precedent at best. A specification limit must be justified by this product's own data, safety qualification and the applicable guideline, not by what another product was allowed.

Check: section choice. The request names 3.2.P.5.5 (Characterisation of Impurities: what each impurity is and where it comes from), but the sample output is a stability trend statement, which belongs in 3.2.P.8.3 (Stability Data) and supports 3.2.P.5.6 (Justification of Specifications).

21. Module by module

Each module gets its own ingestion engine, built like the Module 3 engine, and they all feed one evidence graph (section 8). Module 2 and the label are written **from** that graph, never from scratch.

Module	What it holds	Main sources	Data standards	What the platform does	Hardest check
1 (regional)	Forms (US: 356h for NDA/BLA, 1571 for IND), cover letter, labeling, regulator correspondence, regional documents	RIM system, labeling system, past correspondence	Regional Module 1 specs; FDA Structured Product Labeling (SPL) XML	Fills forms from RIM data, drafts cover letters, drafts the label with every claim linked to evidence	Each label claim backed by Module 5 evidence; regional versions consistent

Module	What it holds	Main sources	Data standards	What the platform does	Hardest check
2 (summaries)	2.3 quality summary, 2.4 nonclinical overview, 2.5 clinical overview, 2.6 nonclinical summaries, 2.7 clinical summary	The evidence graph built from Modules 3–5	ICH M4 structure	Generates tabulated summaries directly from data and drafts overviews with fact placeholders	Every number equals its Module 3/4/5 source; the summaries agree with each other
3 (quality)	CMC	LIMS, Empower, ELN, ERP, CMO certificates	ICH Q guidelines; M4Q(R2) coming	The Module 3 engine (sections 10–20)	Sections 3 and 20
4 (nonclinical)	GLP study reports: pharmacology, pharmacokinetics, toxicology	CRO study reports (PDF), SEND datasets	CDISC SEND, Define-XML	Extracts findings, NOAELs (highest dose with no adverse effect) and exposures; checks report tables against SEND data	Safety margins (animal exposure at NOAEL ÷ human exposure) computed by code and consistent across 2.4, 2.6 and the label
5 (clinical)	Protocols, statistical analysis plans, clinical study reports (CSRs), datasets, integrated safety and efficacy summaries	EDC system, statistical computing environment (SAS/R), safety database, CTMS	CDISC SDTM, ADaM, Define-XML; ICH E3 for CSRs; ICH M11 for protocols	Drafts CSR sections from protocol, analysis plan and output tables; drafts patient safety narratives; independently re-derives key results	Numbers in text = output table = analysis dataset; patient privacy

Plus one cross-cutting step at the very start: the content plan. From the submission type (IND, NDA, BLA, supplement) and region, the platform proposes which documents each module needs, who owns each, and which data feeds it. A regulatory lead approves it, the same way the Module 3 data manifest is approved (section 11).

22. Modules 4 and 5: study data and reports

Clinical and animal data already arrive in strict, machine-readable formats, which makes Modules 4 and 5 better suited to the facts-first design than CMC. The platform must speak these formats natively.

The CDISC standards, in plain words. FDA's binding Guidance on Standardized Study Data requires them for NDAs, ANDAs and certain BLAs, for studies starting from roughly late 2016 onward:

Standard	What it is	Example
SEND	Animal study data in standard tables	Body weights, lab values, microscopic findings per animal
SDTM	Clinical trial data as collected, in standard "domains"	DM (demographics), AE (adverse events), LB (labs), VS (vital signs)
ADaM	Analysis-ready datasets built from SDTM	ADSL (one row per subject), ADAE, ADTTE (time to event)
Define-XML	The data dictionary describing every dataset and variable	Origin and derivation of each variable
Controlled terminology	Fixed vocabularies for values	Standard units, test names

These are submitted in Module 5 (clinical) and Module 4 (nonclinical) with reviewer's guides, and analysis programs for key efficacy tables are submitted too.

The traceability chain the platform walks: case report form → SDTM → ADaM → output table (TLF: tables, listings, figures) → CSR text → Module 2 → label. Every number in the last three must trace back through the first four.

What the platform adds at each link:

1. **Reads numbers from validated outputs**, never computes official results. Efficacy and safety statistics belong to the sponsor's validated SAS or R programs.
2. **Independent re-derivation of key results.** Pharma already checks important outputs by "double programming": a second programmer reproduces each table independently. The platform automates part of this for headline numbers (patient counts, response rates, means) straight from ADaM with its own validated library, and flags any mismatch. It is a check, not a replacement.

3. **CSR drafting by ICH E3 structure.** Methods sections come from the protocol and analysis plan; results sections from the output tables, with fact placeholders.
4. **Patient safety narratives.** Short case histories for deaths, serious and notable adverse events, drafted from the SDTM adverse event, medication and lab records. High volume, highly repetitive, and every fact in them is checkable against the data, so a very good fit.
5. **Module 4 extraction.** NOAELs, exposure values (AUC, Cmax) and target-organ findings come from GLP reports and are checked against SEND. Safety margins are computed by code.

Evidence that this is feasible: Weave Bio and Takeda reported cutting nonclinical summary drafting from about 100 hours to 3, with 95% source fidelity (HIT Consultant, Sept 2026). These are company-reported figures. The 95% also shows the gap this platform targets: the remaining 5% is what deterministic checks should catch.

23. Module 2 summaries and labeling

Module 2 and the label are **views** over the evidence graph. That single design choice is what keeps them consistent with Modules 3 to 5 automatically.

Module 2, part by part:

Part	Built from	How
2.3 Quality Overall Summary	Module 3 facts	The Module 3 engine. Under the draft ICH M4Q(R2), 2.3 becomes the primary assessed quality document (section 29), so this matters more, not less
2.4 Nonclinical Overview	Module 4 facts and claims	Drafted with placeholders; safety margins from code
2.6 Nonclinical Written and Tabulated Summaries	Module 4 facts, SEND	The **tabulated** parts are generated straight from data as tables, with no model in the loop
2.5 Clinical Overview	Module 5 facts and claims	Drafted; benefit-risk conclusions written and signed by named clinical experts

Part	Built from	How
2.7 Clinical Summary (biopharmaceutics, clinical pharmacology, efficacy, safety)	ADaM-based outputs, CSRs	Tables generated from data; text drafted with placeholders

Labeling. The US prescribing information follows a fixed format: Highlights, then 17 numbered sections (1 indications, 2 dosage, 5 warnings, 6 adverse reactions, 12 clinical pharmacology, 14 clinical studies, and so on). FDA also receives it as SPL XML. The platform:

1. Keeps a **label claim traceability matrix**: each label statement linked to its supporting evidence in Modules 2–5. Industry teams maintain this by hand today.
2. Drafts label text from the evidence graph with placeholders, so the adverse reaction table in section 6 can never drift from the safety outputs.
3. Derives regional versions (EU summary of product characteristics, Japan package insert) from the same claim set, with regional wording rules.

Guardrail: label wording is negotiated with the regulator line by line. The platform drafts, links and checks; the labeling team decides. Every negotiated change is captured as a new version of the claim, so the history is never lost.

24. Module 1, multi-region reuse, publishing and lifecycle

Build one global evidence graph with regional overlays, and integrate with existing publishing tools rather than rebuilding them. Publishing is a mature commodity; the platform's value is making sure what gets published is right.

Module 1 is the regional wrapper. US: forms 356h or 1571, cover letter, user fee, patent and financial disclosure information, labeling. EU: application form, product information (SmPC, package leaflet), risk management plan. Japan: Japanese-language administrative documents. The platform fills forms from RIM master data and drafts letters from the content plan.

Multi-region reuse. Modules 2 to 5 are harmonized by ICH, so one evidence graph serves every region. The regional deltas are narrow and listable: Module 1, pharmacopoeia references (USP, Ph. Eur., JP), regional labeling formats, language, and some regional Module 3 content (3.2.R). The platform holds a **core dossier plus regional overlays**, and the consistency engine checks that overlays never contradict the core.

Publishing: integrate, then check. Assembling the eCTD (folder tree, XML backbone, lifecycle operations such as new, replace and delete, hyperlinks, bookmarks) is done well by existing tools: Veeva's submissions publishing, Lorenz docuBridge, Extedo. The platform hands finished documents to them and adds **pre-publishing checks**: PDF rules (embedded fonts, bookmarks, working links), correct document granularity and lifecycle operation, and the regional validation rules a gateway applies automatically before any human reviewer sees the submission.

eCTD v4.0 is arriving region by region (IntuitionLabs summary; several dates come from consultancies, so confirm with each agency):

Region	Status
US (FDA)	Accepted voluntarily for new applications since 16 Sept 2024; no mandatory date announced
EU (EMA)	Optional for new centralized applications from 22 Dec 2025; mandatory reported for 2027
Japan (PMDA)	Reported mandatory from 1 April 2026
Health Canada	Reported voluntary 2026, mandatory 2028

v4.0 gives every document a unique ID that later submissions can **reuse by reference**, tags documents by "context of use" instead of folder position, and uses controlled vocabularies. That suits a graph-based platform: give each evidence-graph document the same identity as its v4.0 document ID, and reuse and lifecycle tracking come almost for free.

25. After submission: questions, commitments and changes

A submission is not the end. Regulator questions, promises made in answers, and years of post-approval changes all draw on the same evidence, so the graph keeps paying off long after the first filing.

Answering regulator questions. FDA sends information requests during review; EMA sends formal lists of questions. The flow:

1. **Intake and classify** each question by topic and the dossier sections it touches.
2. **Retrieve evidence** from the graph, not just text search, so the draft answer cites the exact facts and studies.

3. **Draft** with fact placeholders, route to the right experts, and track deadlines.
4. **Check consistency**: the answer must not contradict what the dossier already says. If it changes a fact, the change propagates (section 8) and the affected sections are queued for an amendment.

Weave Bio already offers question intake and response drafting, and Takeda uses it (HIT Consultant). The difference here is step 4.

Commitments. Answers and approvals create obligations: post-marketing requirements and commitments, promised stability updates, study reports due by a date. Each becomes a tracked item in the graph, linked to the evidence that will fulfil it, with reminders.

Post-approval changes. This was the strongest future idea in section 32, now platform-wide. A proposed change (a new supplier, site, test method or label update) runs through the graph to list every affected product, region and section, and the filing category each region requires: in the US a prior approval supplement, a changes-being-effected supplement or an annual report; in the EU a Type IA, IB or II variation. The platform then drafts the variation package from the updated evidence.

Later scope: periodic safety reports and annual reports reuse the same graph, but they belong to a later phase.

26. Privacy, security and validation with patient data in scope

Module 5 brings patient-level data into the platform, which changes three things: privacy law now applies, a new access rule (blinding) appears, and the redaction agent from the original overview becomes genuinely necessary.

Privacy. Clinical datasets use subject codes instead of names, but under GDPR pseudonymized data is still personal data. Patient narratives, listings and case report forms are the most sensitive. Rules:

- **Minimize what models see.** Most drafting works from aggregated output tables. Patient-level records go to a model only for narratives, and only to a model running inside the sponsor's environment.
- **Purpose-limited access**: per program, per study, per role, all logged.

Blinding firewalls. While a trial is still running, treatment assignments are hidden from most of the sponsor's team so they cannot bias it. A platform that reads trial data must enforce this: unblinded data visible only to authorized roles, and nothing derived from it

leaking into documents other people read. A single leak can compromise a trial.

Redaction becomes a real feature. In section 14, a HIPAA redaction agent made little sense for CMC data. For clinical documents it is needed: CSRs and other clinical documents are published under transparency rules such as EMA Policy 0070, Health Canada's public release of clinical information, and the EU Clinical Trials Regulation. These require anonymizing patients and removing commercially confidential information. The platform can propose redactions with reasons, for a human to approve.

Validation. Everything in sections 18 and 28 still applies: Part 11, the ledger, frozen GEPA prompts, controlled Agent Skills. Two additions:

- **ICH E6(R3)**, the revised Good Clinical Practice guideline adopted by ICH in January 2025, expects computerized systems used for trial data to be fit for purpose and validated in proportion to risk. Anything that touches clinical data must meet it.
- **A credibility file per use.** For each AI use (narrative drafting, CSR drafting, question answering), keep a short record following FDA's draft framework: question, context of use, model risk, tests run, results. Most uses stay in "drafting a submission", but the record proves it.

27. "Why modern AI makes this possible", checked

The "why now" argument is broadly correct: in 2021 this system could not have been built to this standard. A few of the before/after pairs are overstated, and the two strongest drivers are missing.

Capability	Overview: 2021 → 2026	Verdict
Context window	2k–8k tokens → 1M+	True. Opus 4.6, Opus/Sonnet 5.5 and DeepSeek V4 all offer 1M. Caveat: long context is not perfect recall (Opus 4.6 scored 76% on a long-context retrieval test, so it still misses), so exact values still come from the fact store

Capability	Overview: 2021 → 2026	Verdict
PDF parsing	OCR + regex → vision-language models	Overstated gap. Layout-aware OCR already existed by 2021 (AWS Textract, Azure Form Recognizer, LayoutLM). The real change is **semantic** understanding of messy layouts
Integration	Point-to-point APIs → MCP	True. MCP only appeared in November 2024. The adapters still have to be written
Retrieval	Naive vector search → hybrid Graph RAG	Partly. Hybrid keyword + vector search and knowledge graphs predate 2021. What is new is LLMs making graphs easy to build and query (for example text-to-Cypher)
Arithmetic	Probabilistic → code sandboxes	True in spirit. Code always computed exactly; what changed is models reliably writing code and calling tools
Workflow	"Linear chains (LangChain 2021)" → LangGraph	Direction true, date wrong: LangChain launched in October 2022 and LangGraph in January 2024

Two drivers the overview misses, and they are the strongest:

1. **Reliable structured output and tool calling.** Since 2024 models can be constrained to emit valid JSON matching a schema and to call typed tools. This is what makes "model outputs a fact with a pointer" and "model calls `shelf_life(fact_ids)`" dependable, far more than raw context size.
2. **Regulators started writing AI rules.** FDA's draft AI credibility guidance (January 2025), the FDA–EMA Good AI Practice principles (January 2026) and the EU's draft GMP Annex 22 (July 2025) give sponsors a framework for defending AI-assisted work. Before them, no QA head would sign off. Regulatory clarity is a precondition for buyers, not a footnote.

28. Latest GenAI and agentic technologies to add

Ten technologies from 2025–2026 earn a place, and each one strengthens a box that already exists (see the section 10 diagram) rather than adding a new layer. **GEPA is the strongest addition**: it improves every prompt in the system automatically, using the checkers this project already has. **JEPA is real and important research, but it does not fit the core of this project**; it has two narrow, optional uses, described below.

GEPA: self-improving prompts

What it is. GEPA (Genetic-Pareto) is a prompt optimizer from UC Berkeley, Stanford, Databricks, MIT and others, accepted as an ICLR 2026 oral paper. It runs your pipeline on example inputs, has a strong model **read the traces and the error messages in plain language**, and writes improved prompts. It keeps a Pareto frontier of candidate prompts (each best on different examples) instead of polishing a single one, and can merge lessons from different candidates. In the paper it beat GRPO, a reinforcement-learning method that changes model weights, by about 6% on average (up to 20%) while using up to 35× fewer runs, and beat the older MIPROv2 optimizer by over 10%. It ships as `dspy.GEPA` in the DSPy library.

Why it fits this project so well. GEPA learns best from rich written feedback, and this project already produces exactly that from its deterministic checkers. When the CoA extractor gets a row wrong, the cross-check can say: *"Impurity B, Month 12: extracted 0.14, expected 0.014; decimal shifted; the footnote says results are in %."* When the writer overclaims, the validator can say: *"Sentence 3 says all batches met specification, but batch 2024-07 failed dissolution."* GEPA reads hundreds of such messages and rewrites the instructions to stop those mistakes.

Where it runs: offline only. GEPA tunes prompts during development. The winning prompt is then frozen, regression-tested on the golden set, version-pinned and logged in the ledger like any other change (section 30, model change control). Production never edits its own prompts, which keeps validation intact.

```
import dspy

class ExtractCoA(dspy.Signature):
    """Extract every result row from a Certificate of Analysis page."""
    page_text: str = dspy.InputField()
    rows: list[dict] = dspy.OutputField(desc="analyte, value, unit, limit")

def metric(gold, pred, trace=None, pred_name=None, pred_trace=None):
    errors = compare_rows(gold.rows, pred.rows)      # deterministic checker
    score = 1 - len(errors) / len(gold.rows)
    return dspy.Prediction(score=score, feedback="\n".join(errors) or "All
rows correct.")

optimizer = dspy.GEPA(metric=metric, auto="light",
                      reflection_lm=dspy.LM("anthropic/claude-opus-5-5"))
better = optimizer.compile(dspy.Predict(ExtractCoA), trainset=train,
valset=val)
```

Watch out: it can overfit its training examples, so always judge the result on a held-out set it never saw. The synthetic CoA generator from section 36 gives you the hundreds of labeled examples it needs for free.

JEPA: world models, an honest fit check

What it is. JEPA (Joint Embedding Predictive Architecture) is Yann LeCun's alternative to generative models. A normal model learns by predicting the missing raw data: the next word, the missing pixels. JEPA predicts the missing part's **meaning** (its embedding) instead, which pushes it to learn structure and ignore noise. The family so far:

- **I-JEPA** (images, 2023).
- V-JEPA 2 (video, June 2025; 300M–1B parameters, MIT license). Its action-conditioned version, V-JEPA 2-AC, plans robot arm movements; V-JEPA 2.1 followed in March 2026.
- LLM-JEPA (September 2025, Huang, LeCun, Balestriero): adds a JEPA loss to language-model training using **two views of the same knowledge**, such as a text description and its code. It improved natural-language-to-regex, text-to-SQL and GSM8K math at 1B–8B model sizes, and resisted overfitting better than plain fine-tuning.
- **AMI Labs**, LeCun's new company, raised a $1.03 billion seed round in March 2026. Its targets are robotics, driving, wearables and industrial processes, and it had no product yet as of September 2026.

Why it is not in the core. This project's core job is reading documents exactly and checking numbers. JEPA models do not generate text and do not read documents better than today's OCR plus vision models, and no production JEPA model for documents exists. Putting one in the main path would add risk without adding accuracy.

Two places it could genuinely help (optional research track, after the MVP):

1. **LLM-JEPA for a small local extractor.** This project naturally has two views of the same knowledge: the text of a scanned table row and its verified fact JSON, or a narrative sentence and the facts it states. Fine-tuning a ~1B local model with the LLM-JEPA loss on those pairs might make extraction more robust than plain fine-tuning. It is testable with the synthetic data.
2. **Self-supervised anomaly flags on raw signals.** A JEPA-style encoder trained on chromatogram traces or stability curves could flag unusual ones for human review, as an early out-of-trend warning. It would **never** be the source of a reported value, which keeps it consistent with the rule in section 6. This also points toward the process-analytics future scope (section 32), the same industrial area AMI Labs targets.

Verdict: a strong thesis chapter or paper, not an MVP component.

The other nine technologies, and where each plugs in

Technology (when)	What it is	Plugs into (section 10 box)	What it adds	Watch out
Structured outputs (Claude, Nov 2025; now standard)	The schema is compiled into a grammar, so output always matches it	Scans: OCR + VLM; Writer model	Every fact proposal is valid JSON with the right fields; no parse failures or retries	Schemas cannot express numeric min/max, so ranges are still checked in code; cannot be combined with Citations in one call
Citations (Claude)	Answers point to exact passages and page numbers in supplied documents	Rules check	Every regulatory statement quotes the exact ICH passage and page it relies on	Works only on PDFs with a text layer (fine for ICH PDFs); run as a separate call from structured outputs

Technology (when)	What it is	Plugs into (section 10 box)	What it adds	Watch out
Local document VLMs: PaddleOCR-VL-1.6, Granite-Docling, DeepSeek-OCR, dots.ocr	Small open models that convert page images to structured text and tables	Scans: OCR + VLM	Run inside the VPC on one GPU. PaddleOCR-VL-1.6: 0.9B parameters, Apache-2.0, 96.33% on OmniDocBench v1.6; Granite-Docling: 258M, outputs Docling JSON	Keep the two-route cross-check; confirm each model's bounding-box output before relying on it
ColPali / ColQwen visual retrieval	Searches page **images** directly with a vision model and late-interaction matching, no OCR needed	Knowledge layer	Finds the right scanned page ("which CoAs mention Impurity B?") even when OCR misread the text	Larger multi-vector index; use it to find pages, never to read values
MCP Tasks and elicitation (Nov 2025) and the 2026-07-28 spec	Long jobs return a task handle; a server can pause and ask the human for input; W3C trace context in calls	MCP servers	Long LIMS and Empower pulls run asynchronously; the server can ask the reviewer to approve the data manifest mid-call; every call carries a trace ID into the audit trail	The 2026-07-28 release is breaking (stateless core, Tasks moved to an extension): pin one spec version
Advanced tool use (Claude, Nov 2025): Tool Search, Programmatic Tool Calling, Tool Use Examples	Load tools on demand; the model writes a short script that calls tools in a sandbox; example calls attached to tool definitions	Orchestrator; Statistics	The model composes calls such as `shelf_life(fact_ids)` in a script, so raw values never pass through its context (37% fewer tokens reported). Tool examples teach exact fact-ID formats (72% → 90% parameter accuracy reported)	Log every generated call script in the ledger
Agent Skills (Oct 2025; open standard Dec 2025)	Versioned folders of instructions, templates and scripts that an agent loads when relevant	Writer model; Rules check	One skill per section type, e.g. a 3.2.P.8.3 stability skill with its template, required tables, phrasing rules and examples. Regulatory experts can review it like an SOP	Treat each skill change as a controlled document change

Technology (when)	What it is	Plugs into (section 10 box)	What it adds	Watch out
OpenTelemetry GenAI conventions	Standard trace format for model calls, agent steps, tool calls, retrieval and evaluations	Audit ledger	Every step becomes a standard trace that can go to an observability tool (such as Langfuse) and be hashed into the ledger	Still "Development" status as of July 2026: pin the version
A2A protocol (v1.0, Linux Foundation)	Standard for agents of different companies to talk to each other; complements MCP	Next to Veeva Vault	Later: hand a signed section to Veeva's or a CRO's agents through a standard. 150+ organizations; built into Azure AI Foundry and Amazon Bedrock AgentCore	Not needed for the MVP

Considered and left out

- **Reinforcement-learning fine-tuning (GRPO and similar):** changes model weights, so every run means re-validation. GEPA got bigger gains with prompts alone in its own paper.
- **Self-organizing agent swarms:** unpredictable paths cannot be validated (section 14).
- **Long-term agent memory across filings:** risks carrying one product's facts into another product's dossier. Each run starts only from its approved data manifest.
- **Computer-use agents for CMO portals:** maybe later, only as a supervised, read-only download helper. Never typing values.
- **JEPA in the core path:** see above.

What to add to the student build (section 36)

1. **GEPA** on the CoA extractor and the writer, scored by your own checkers. The biggest win for the least work.
2. **PaddleOCR-VL or Granite-Docling** locally instead of a cloud OCR service.
3. **Structured outputs** for fact extraction and **Citations** for ICH quotes.
4. **OpenTelemetry traces into Langfuse**, hashed into the ledger.
5. **One Agent Skill** for 3.2.P.8.3 stability writing.

6. Optional research chapter: **LLM-JEPA** fine-tuning of a ~1B model on text ↔ fact-JSON pairs from the synthetic data, compared with plain fine-tuning on extraction accuracy.

Updated one-breath pitch: "…every number comes from a verified fact store, and the prompts that read the documents are optimized with GEPA against the system's own deterministic checkers."

29. The structured-data future

Regulators are moving from documents toward structured data, one standard at a time. The strategic bet: **store evidence as structured data and render documents as views of it.** Document-first tools will have to be rebuilt as these standards land; an evidence-graph platform simply adds another output.

Standard	What it does	Status	What the platform does with it
ICH M11 (CeSHarP)	A structured, machine-readable clinical protocol: guideline, template and technical specification with defined data elements	Adopted by ICH 19 Nov 2025; FDA final guidance 22 May 2026; in effect in the EU from 11 June 2026; not mandatory (Sakara Digital)	Reads M11 protocols as data, so CSR methods sections, 2.7 descriptions and trial-registry entries fill themselves
CDISC USDM	The data model for study definitions, built with TransCelerate's Digital Data Flow and aligned with M11	In use; CDISC standard	The exchange format for protocol data into the graph
ICH M4Q(R2)	Restructured quality modules: 2.3 becomes the primary assessed document; Module 3 the supporting repository; most parts follow Description, Manufacture, Control, Storage	Step 2 draft May 2025; consultation closed Oct 2025; Step 4 targeted June 2027 (EUREG)	The Module 3 fact store maps onto it; it is designed to prepare for a forthcoming ICH guideline on structured quality submissions

Standard	What it does	Status	What the platform does with it
FDA PQ/CMC	FDA's program to standardize quality data elements	Ongoing	Same target as M4Q(R2): CMC facts submitted as data
ISO IDMP	Standard identifiers for medicinal products and substances	Implemented through EMA's master-data services	Product master data in the graph, shared with Module 1 and labeling
eCTD v4.0	Message-based submission with reusable document IDs	Rolling out (section 24)	Graph document IDs = v4.0 IDs

Practical rule for the build: every evidence item gets a stable ID and machine-readable fields from day one, even while outputs are still Word and PDF. That costs little now and turns each new standard into an export, not a rewrite.

30. Technical challenges

The overview's five challenges are real, but it misses the hardest one: **how do you validate a system whose AI parts can answer differently each run?** The answer shapes the whole architecture: validate the deterministic controls around the model, not the model itself.

The overview's five, judged:

Challenge	Overview's mitigation	Verdict
1. Zero hallucination tolerance	Models write Python for math; a hard block stops unverified numbers	Right idea. Strengthen with fact placeholders and a claim validator (section 6). Hallucination is not only numbers: a wrong batch ID, method name, or "no new impurities were observed" are claims too

Challenge	Overview's mitigation	Verdict
2. Part 11 compliance	Immutable Postgres ledger with SHA-256 chains	Necessary, not sufficient; close the three holes in section 18. The real Part 11 challenge is validation (below)
3. Data sovereignty	GovCloud, Nitro Enclaves, local vLLM	Right goal, inconsistent choices (Pinecone, Aura, OpenAI embeddings, Opus 4.6 outside GovCloud). Add contractual zero-data-retention with any model provider
4. Ingestion latency	Celery + Redis Streams, GPU batching	Fine; add idempotency keys (section 19)
5. Token cost	Local model for preliminary work, Claude for synthesis	Fine; add prompt caching, and extract facts once so no step re-sends a million tokens

What the overview misses:

1. **Validating non-deterministic AI.** FDA's January 2025 draft guidance defines a 7-step credibility framework: define the question, define the context of use, assess model risk, plan, execute, document, decide adequacy. Model risk = **model influence** (does a human check the output?) × **decision consequence** (what happens if it is wrong?). Notably, the draft excludes AI used "to streamline operations (e.g., drafting a regulatory submission)" when it does not affect patient safety or data reliability. This project's design fits that well: the AI drafts and extracts, while every number is produced by validated code and verified by a human, so the AI's influence on the evidence is low. Validation then targets the controls: extraction cross-checks, the statistics library, the validator, and the review workflow.
2. **Ground truth for testing.** You need real scanned CoAs and real approved Module 3 sections to measure accuracy. Module 3 content is confidential; FDA's public review packages contain reviewers' summaries, not the dossiers. Synthetic documents miss real scan chaos. A design-partner company is the only realistic source.
3. **Shared vocabulary.** The same impurity, test or unit is named differently across sites and CMOs. The system needs a controlled vocabulary with synonyms, maintained by people. FDA's own PQ/CMC initiative to standardize quality data elements points the same way.

4. **Model change control.** Pin exact model versions; any change of model, prompt or library version triggers regression tests on a golden set before release.
5. **Late data.** A new stability pull arrives after drafting. With fact lineage, the system finds every sentence and table that depends on the changed data and re-runs only those.
6. **Prompt injection.** A document can contain text that tries to instruct the model. Treat all document text as data, restrict tools to read-only, and never let document content change the workflow.
7. **Human factors.** Writers must trust the tool without over-trusting it (section 17), and it must fit into Word and Veeva, where they already work.
8. **Liability.** The sponsor's QA stays responsible for what is submitted. Contracts and the product's claims must say so plainly: the tool assists and proves; it does not certify.
9. **Format transitions.** FDA has accepted eCTD v4.0 voluntarily since 16 September 2024, with no mandatory date yet. Output documents must fit both v3.2.2 and v4.0 publishing.

31. Roadmap and exit criteria

The overview plans four 8-week phases, 32 weeks from empty cloud account to a validated production pilot on a real IND or BLA. Building a thin working slice in that time is plausible; connecting three live enterprise systems, completing formal validation and piloting a full Module 3 in 32 weeks is not. The order is also backwards: the review screen and validation come last, when they should shape everything from week one.

The overview's phases and exit criteria, judged:

Phase (weeks)	Delivers	Exit criterion	Problem	Better criterion
1. Foundation (1–8)	GovCloud VPC, MCP servers for Veeva, LIMS, ELN, audit ledger	Queries across all three in under 500 ms	Speed is not the risk; access to a customer's Veeva and LIMS alone depends on their IT and security review	Every record of a reference study retrieved; counts and checksums match a manual export

Phase (weeks)	Delivers	Exit criterion	Problem	Better criterion
2. Ingestion (9–16)	VLM table extraction, Neo4j schema, Pinecone hybrid search	>99.5% extraction accuracy on 2,000 CoAs	Undefined unit; still ~1 error per 200-value CoA	Zero **undetected** errors on a held-out set: every wrong value flagged or caught; plus reviewer minutes per document
3. Agents (17–24)	LangGraph supervisor, Python sandbox, Claude + R1	Zero calculation errors on 500 stability matrices	Trivially true for deterministic code	Library outputs match an independent reference (JMP/SAS) on edge cases: missing points, "<LOQ", rounding ties, non-poolable batches
4. UX + validation (25–32)	Dual-pane UI, GAMP 5 validation, pilot filing	Full Module 3 in under 48 hours with 100% human sign-off	Machines may finish in 48 h; humans verifying thousands of values will not	End-to-end cycle time **including review** vs the manual baseline, and errors caught in a seeded-error study

Why the order is backwards:

- **Validation is a lifecycle, not a final phase.** Under GAMP 5 (ISPE, 2nd edition 2022), user requirements, risk assessment and the test strategy are written at the start and drive the design. Formal validation of a new GMP system commonly takes months on its own. FDA's Computer Software Assurance guidance (final, September 2025) promotes risk-based, lighter-weight testing; it was written for medical-device makers, but pharma widely borrows its approach.

- **The review screen is the product.** Users judge the system by it, and its needs (fact IDs, source boxes, review states) define the data model underneath. Build a thin version first.

A more realistic roadmap, as vertical slices:

1. **Setup:** sign a design partner; write user requirements and a risk assessment; build a golden test set from their real, anonymized documents; pick one section.
2. **Slice 1, structured data only:** 3.2.P.8.3 stability tables from LIMS exports → fact store → statistics library → table + narrative with fact placeholders → review screen → ledger. No scans, no live connectors yet.
3. **Slice 2, scans:** add CoA parsing with the OCR + VLM cross-check, and 3.2.P.5.4 batch analyses.
4. **Slice 3, consistency:** the cross-section checker (3.2.S vs 3.2.P vs Module 2.3) and live MCP connectors.
5. **Shadow mode:** run on a real filing **in parallel** with the manual process and compare errors found and hours spent, while formal validation runs.

Shadow mode is the step that turns a demo into evidence a QA head will accept.

The platform roadmap beyond Module 3

Build outward from Module 3, one module at a time, with the evidence graph from day one:

1. **Module 3 + 2.3** (sections 10–20). Evidence graph and ledger built here, even though only CMC uses them at first.
2. **Module 4 + 2.4/2.6.** SEND data and GLP reports. The smallest data step; Weave's own entry point (nonclinical summaries) shows the demand.
3. **Module 5 + 2.5/2.7.** CDISC datasets, output tables, CSR drafting, patient narratives, re-derivation of key results.
4. **Cross-module consistency and labeling.** Consistency rules across modules, label claim matrix, change propagation.
5. **Module 1, multiple regions, pre-publishing checks**, with hand-off to existing publishing tools.
6. **After submission.** Regulator questions, commitments, post-approval change packages.

Each phase ends with a seeded-error study and a shadow run against a real team's manual work, as described above.

32. Future scope and market

Of the three expansion ideas, post-approval impact analysis is the strongest and closed-loop bioreactor control the weakest. The $8.4B market figure has no source and is probably inflated for software alone.

The three expansions, judged:

Idea	What it means	Strength	Problem	Verdict
Closed-loop bioreactor optimization	When stability data trends badly, recommend changes to feed strategy or harvest timing, connected to the MES	Big value if it worked	Stability results arrive months or years after the batch was made, too late to steer it. Process changes need change control and often prior regulator approval. Draft EU Annex 22 says LLMs should not be used in critical GMP applications	Weakest. A different product (process analytics) needing validated static ML models with human approval, not LLMs
Multi-agency translation (FDA → EMA, PMDA, NMPA)	Re-format a validated US Module 3 for other regions	Module 3 is already harmonized (ICH M4Q), so the fact store can be reused almost as is	The real differences sit in Module 1, 3.2.R, pharmacopoeia references (USP vs Ph. Eur. vs JP), and language (Japanese, Chinese). Needs EU and Asia deployments, which GovCloud blocks	Natural extension

Idea	What it means	Strength	Problem	Verdict
Post-approval impact agents	A change is proposed (swap a filter supplier); find every affected product license worldwide and draft the variation filings	Recurring, continuous work; the graph's impact analysis (section 13) does exactly this; aligns with ICH Q12 lifecycle management	Registration data lives in RIM systems, where Veeva is the incumbent	Strongest. Likely the best business, possibly via partnership

Check: the market size. The overview's "$8.4 billion addressable market across regulatory management software and services" has no source. For comparison, Grand View Research projects the regulatory information management software market at $5.11 billion by 2033 (9.1% annual growth from 2026). The $8.4B probably adds outsourced regulatory services. A credible pitch builds the number **bottom-up** instead: target companies × CMC submissions per company per year (INDs, NDAs/BLAs, supplements) × price per submission or per product. Each input can be researched and defended.

Who buys, and how:

- **Economic buyer:** head of Regulatory CMC or VP Regulatory Affairs.
- **Must approve:** QA (validation), IT security (the deployment), sometimes legal (data rights with CMOs).
- **Channel:** small biotechs outsource regulatory writing to CROs and consultancies, which can be partners. Weave Bio's work with Parexel shows the pattern.
- **Beachhead:** mid-size biotechs that file often and rely heavily on CMOs. They have the most scanned CoAs, feel the pain most, and decide faster than big pharma.
- **Sales cycle:** long. Security review, validation and a shadow-mode pilot come before any production use.

Competitors are listed in section 5. The positioning that survives them: verification and provenance first, drafting second.

33. How this differs from Weave Bio and the others

Weave Bio is **drafting-first with traceability**: it writes documents and links each sentence back to a source paragraph. This platform is **evidence-first**: it builds a verified graph of facts and claims, and documents are views of it. That is a different architecture, not a copy, and it targets the part buyers worry about most: proving the dossier is right and keeping it right.

Against Weave Bio, using only what it has published:

Capability	Weave Bio (public information)	This platform
Scope	IND drafting incl. nonclinical summaries; NDA workflow launched with Parexel (April 2026); regulator question workflows; Takeda across 14 programs	All modules plus labeling, multiple regions and post-approval lifecycle
Traceability	Sentence-level tracing to sources; “always-current links” and Data Tags reuse values across a document set	Each value mapped to its exact cell, dataset row or page box; claims typed and linked to evidence sets
Number checking	Automated verification of critical data and 95% source fidelity reported, yet its own study found misread values (section 34)	Deterministic validator: every digit must match a fact; key results re-derived independently from the datasets
Cross-module consistency	AutoReview flags inconsistencies, omissions and conflicts; how it works and how far it reaches across modules is not published	Explicit consistency rules across modules, label and regions, with change propagation
Structured data	Not described beyond structured eCTD outputs and DOCX export	Native SDTM, ADaM, SEND, Define-XML; ready for M11/USDM, M4Q(R2) and eCTD v4.0 IDs
Compliance evidence	Audit trail of every edit, comment and version; role-based permissions; SOC 2 Type II	Part 11 hash-chained ledger, frozen and versioned prompts, a credibility file per AI use

Capability	Weave Bio (public information)	This platform
Openness	Pulls source files from Veeva and resyncs them; other integrations not described	MCP connectors and A2A hand-offs; works alongside Veeva

The others, in one line each: Celegence and SyncIQ focus on CMC and Module 2.3 (SyncIQ sells cross-module 2.3/3 consistency, a narrow version of section 8). Peer AI uses specialized agents across preclinical, clinical and CMC documents. Veeva owns the document platform and plans agentic authoring for late 2027. All of them lead with faster drafting.

Honest caveat: Weave has paying customers, real data and a CRO partner; this is a design. The differences in the table are claims until a benchmark proves them. The benchmark to build is a **seeded-error study**: plant known mistakes across modules (a wrong number in 2.7.3, a dose that differs between the label and a CSR, an outdated shelf life in 2.3) and measure how many each approach catches. Catching nearly all of them is the pitch.

One-line positioning: "Other tools write your dossier faster. This one proves every claim in it, across every module and region, and keeps it proven as the data changes."

34. Product experience: learning from Weave Bio, and going further

What this is based on. Weave Bio publishes no screenshots, public demo or user reviews; its site has a product video and written descriptions of each workflow. This section works from Weave's product and security pages, plus the AutoIND study that Weave and Takeda authors published (arXiv 2509.09738), which measured how much of the AI's drafts an expert assessor had to fix. Read every statement about Weave's screens as "what Weave describes", not "what was seen".

The strategy in one line: match everything Weave has made standard, then win where its own study shows gaps: proof you can see, review by exception, and drafts that need less rework.

What Weave Bio does well (now table stakes)

Pattern	What Weave does	Why it works
Shaped like the job	Submission Builder follows five steps: author, assemble the dossier, review together, verify sources, export	Writers see their own process, so there is little to learn
Templates, not a chat box	Configurable templates with variables, instructions and tables, “grounded in source context” instead of “disconnected prompting”	The structure comes out right every time, and a team sets its house style once
Organised by lifecycle stage	Candidate → Pre-Clinical → Clinical → Market → Post-Market, across Program Planner, Submission Builder and HAQ Manager	People find work by where the drug is, which is how regulatory teams think
Sources one click away	Sentence-level tracing, “always-current links”, Data Tags for reused values, literature citations	A reviewer checks a claim without leaving the page
An automatic second reader	AutoReview flags inconsistencies, omissions and conflicts	Problems surface before a person spends review time
Collaboration in one place	Shared workspace with comments, redlines, section status, version restore, an audit trail of every edit, and role-based permissions	Replaces email chains and files named final_v7
Word as the exit	Customisable DOCX export; source files pulled from Veeva keep their folder structure and resync	Fits the tools publishers and reviewers already use

Pattern	What Weave does	Why it works
Agency questions as a tracker	HAQ Manager extracts each question, assigns owner, reviewer, approver and due date, finds similar past questions, and drafts only from sources the user explicitly selects	A stressful deadline becomes a visible queue, and explicit source selection stops one program's data leaking into another
Trust stated up front	SOC 2 Type II, zero data retention with model providers, per-program data rooms, model updates gated by a regression suite	Answers the first questions every pharma IT and QA team asks

Do not try to beat Weave on these. A product without them will not get past a first demo, so build them to the same standard and spend the extra effort on what follows.

Where Weave's own study shows room to do better

In the AutoIND study (two INDs, one blinded assessor), drafting time fell by about 97%, from roughly 100 hours to 3.7 and 2.6 hours, with no critical regulatory errors. But quality scored 69.6% and 77.9%, and the authors write that 10 to 25% of generated content needed revision. The gaps are specific, and each one is a design problem with a design answer.

Finding	Evidence from the study	Our design answer
Drafts too long	AI documents were 3 to 5 times longer than human-written ones	A live length budget per section, set from expert examples and shown while writing
Methods over results	Emphasis scored 6.7% and 37% in toxicology	Results-first templates, and an emphasis view that tints methods and results differently
Required content missing	Toxicology completeness 32% and 59%; 35% of summaries missed essential study-design elements; GLP dose-formulation analysis was omitted every time	A required-elements checklist per section (from ICH M4, E3 and the content plan) that blocks "ready for review" until each item is present or marked not applicable

Finding	Evidence from the study	Our design answer
Structure and AI voice	Of 156 clarity comments, 32% were structural and 28% were AI-style phrasing	The outline is approved before any prose is written, and a house-style checker flags AI-style phrasing
Wrong numbers	A wrong onset time (11 h instead of 7 h); NOAELs assigned to in-vitro studies	Every number is a fact chip with a verification state, and a mismatch blocks sign-off (section 6)
Slow to start	A rival's comparison says onboarding takes weeks of data-room setup (a biased source: it was written by a competitor's founder)	First useful output on day one (principle 7)
Hard to try	Enterprise sales only, no public pricing	A free, self-serve consistency check (below)

Nine principles for our product

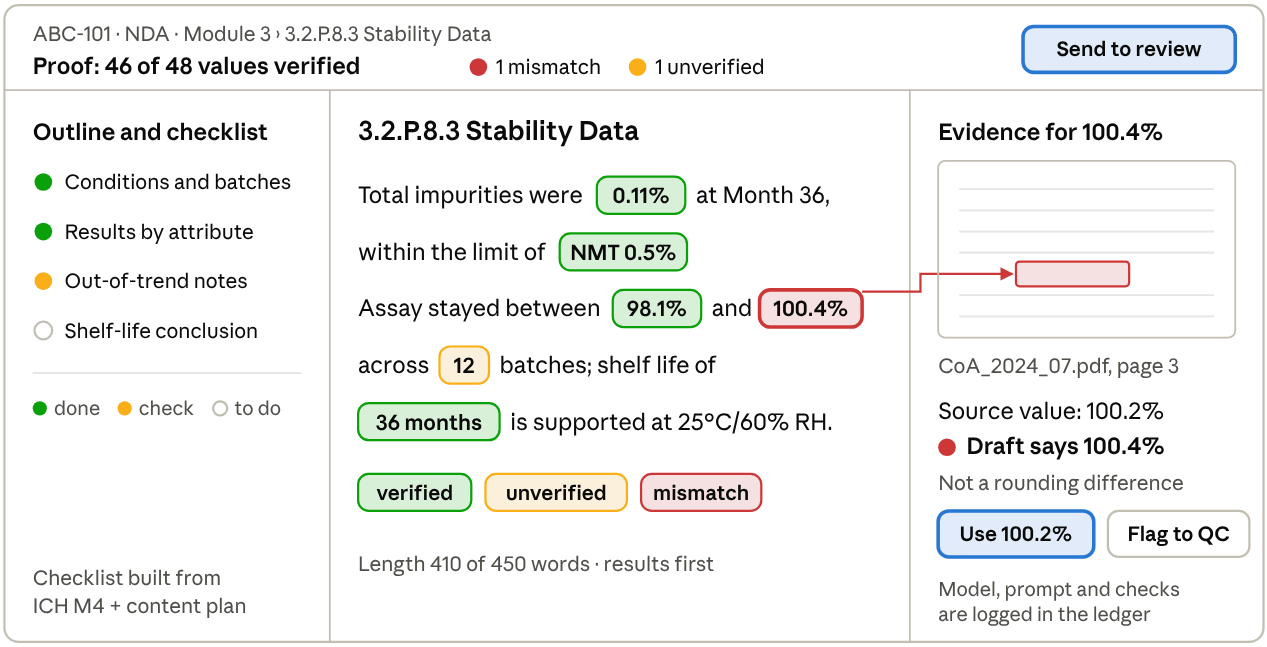
1. **Show proof, do not ask for trust.** Every number and claim carries a visible state: verified, derived, human-confirmed, unverified or mismatch. Each section shows a proof score ("46 of 48 values verified"), so a reviewer sees the risk before reading a word.
2. **Review by exception.** Reviewers work from a queue ranked by risk: mismatches first, then unverified values, then new claims. Verified, unchanged paragraphs are collapsed. The whole queue works from the keyboard.
3. **Outline first, prose second.** The required-elements checklist comes from the guideline and the content plan. The writer approves the outline before drafting, which removes most structural edits before they happen.
4. **Drafts that need less rework.** Length budgets, results-first templates, and prompts tuned against expert edits with GEPA (section 28). Every human edit is kept as a signal for the next round of tuning.
5. **See the consequence of a change.** Changing a fact (a new stability time point, a corrected batch count) first shows every section, table and module it touches, through the evidence graph (section 8). Nothing updates until someone approves the list.

6. **Meet writers in Word.** A Word add-in shows the same fact chips and evidence panel, so writers who never leave Word still get proof and checks.
7. **First value in a day.** Connect a folder or Veeva vault, auto-classify documents into the eCTD tree, generate a content plan, and return a consistency report on the first day, before any drafting.
8. **Compliance that feels normal.** Part 11 signatures state their meaning ("approved as author"), re-authentication happens only at signature, the audit trail reads as a plain-language timeline, and every source carries a coloured program badge so reuse across programs is visible at once.
9. **One screen per role.** The writer, reviewer, CMC lead, regulatory lead and publisher each start on the view their job needs, not one generic dashboard.

The section workspace

The writer sees proof while writing: every number is a chip with its state

Wireframe of the section workspace (illustrative values)



Section workspace wireframe · outline, draft with fact chips, evidence

The writer drafts in the centre, where every number is a chip coloured by its state. Clicking the red 100.4% chip opens its evidence on the right: the source page with the value boxed, the source value (100.2%), and one-click fixes. The checklist on the left

shows what the section still needs, and the proof score in the top bar is what the reviewer sees in their queue. This screen is where sections 6, 8 and 17 become something a person touches.

Six key screens

Screen	Main user	What it shows
Dossier cockpit	Regulatory lead	Every module and section with status, proof score, owner and due date, blockers first
Section workspace	Writer	Outline and checklist, draft with fact chips, evidence panel (drawn above)
Review queue	Reviewer, QA	Risk-ranked items across sections; accept, fix or send back with a reason
Consistency map	CMC and regulatory leads	Conflicts across modules from the evidence graph, such as a shelf life that differs between Module 2.3 and 3.2.P.8
Question board	Regulatory lead	Agency questions as cards with owner, due date, linked evidence and draft answer
Onboarding wizard	Admin	Connect sources, map folders to the eCTD tree, set roles, run the first consistency check

Using the experience to win customers

Weave sells to enterprises through demos and pilots. We can add a self-serve door: a free consistency check where a team uploads two documents, for example a Module 2.3 summary and a 3.2.P.8 stability report, and gets back every mismatch with page references. Files are processed in isolation and deleted after the report. It shows the core value, proof, in the first few minutes, and every report is a lead for the full platform.

How we will measure it

Metric	Our target	Why it matters
Time to first useful output	Under one day from connecting sources	Tests principle 7
Reviewer minutes per section	Falls with every release	Tests review by exception
Rework rate	Below the 10 to 25% in Weave's study	Share of generated text that reviewers change
Length against expert examples	Within 20% either way	The study found drafts 3 to 5 times too long
Seeded-error catch rate	100% of planted wrong numbers	Proves the fact checks work
System Usability Scale (SUS)	Above 70 in pilot tests	A standard 10-question usability score; 68 is average

35. Scorecard: every issue in the overview, with its fix

43 issues: 8 high, 21 medium, 14 low. The high ones cluster in two places: numbers that still pass through the model, and the pharmaceutical science in the worked example. Section numbers in brackets point to the full explanation.

#	Severity	Topic	Overview says	Fix (section)
1	High	Math script	The model types data values into the Python script	Load values by fact ID; the model types none (6, 16)
2	High	Provenance	The model emits the provenance JSON (page, box)	The system attaches provenance from the fact record (6)
3	High	OOT test	Fits the line through the point it tests; flags only at 0.164% or more	Prediction interval from earlier data flags above ~0.124% (16)

#	Severity	Topic	Overview says	Fix (section)
4	High	Sample output	"Month 36 under accelerated conditions"	Accelerated runs 6 months; Month 36 is long-term 25°C/60% RH (20)
5	High	Sample output	Total degradation products compared with a Q3B threshold	Q3B thresholds are per individual product; totals follow the product's own spec (20)
6	High	Guideline choice	ICH Q3B applied in a BLA story	Q3A/Q3B exclude biologics; use Q6B/Q5C, chosen by modality (2, 20)
7	High	Sovereignty	Pinecone Serverless, Neo4j Aura, OpenAI embeddings in a "single-tenant VPC"	Self-hosted vector, graph and embedding models inside the VPC (13)
8	High	Sovereignty	Claude Opus 4.6 inside GovCloud	Not on AWS's GovCloud Bedrock list; pick a model available inside the boundary (15)
9	Medium	Sample output	Q3B 0.20% threshold with no daily dose	Compute the threshold from maximum daily dose (20)
10	Medium	Data sanity	Batch 2024-12 has Month-36 data	Impossible before Dec 2027; add a date rule (20)
11	Medium	Business impact	One decimal error triggers Refuse-to-File	RTF is for incomplete applications; expect an IR or CRL questions (3)
12	Medium	Agents	HIPAA / PII redaction agent	CMC rarely holds patient data; reframe as confidentiality redaction (14)

#	Severity	Topic	Overview says	Fix (section)
13	Medium	Sandbox	"PyChain WASM" runtime	Does not exist; Pyodide, gVisor or Firecracker plus a validated statistics library (16)
14	Medium	Models	DeepSeek-R1 does "preliminary data parsing"	R1 is text-only; it cannot read scans (15)
15	Medium	Models	R1 on "isolated H100 nodes"	~700 GB at FP8 exceeds one 8×H100 node; multi-node or a smaller model (15)
16	Medium	Models	DeepSeek-R1 as the local model	Outdated (V4 shipped Apr 2026) and flagged in security reviews; consider gpt-oss, Llama, Mistral, Granite (15)
17	Medium	Deployment	AWS GovCloud as the default	Commercial region per customer; GovCloud only for government programs (19)
18	Medium	Parsing	VLM reads peak areas off chromatogram charts	Take numbers from Empower or the printed results table (12)
19	Medium	Parsing	VLM-generated bounding boxes as audit provenance	Boxes from the OCR layer, cross-checked with the VLM (12)
20	Medium	Long context	1M tokens removes chunking; ingest 50,000 pages	1M tokens is ~1,500–3,000 pages; use table-aware chunking and the fact store (13, 15)
21	Medium	Math	"100% mathematical accuracy"	True of arithmetic only, not of inputs, methods or rounding (16)

#	Severity	Topic	Overview says	Fix (section)
22	Medium	Agents	Compliance Audit as an agent	Logging must be automatic at framework level (14)
23	Medium	Agents	No consistency checker, no writer node	Add both; cross-section mismatches are the top error source (14)
24	Medium	Regulatory logic	Limits justified by other products' past filings	Justify with this product's data, safety and guidelines (20)
25	Medium	Exit criteria	500 ms latency, >99.5% accuracy, zero calc errors, 48 h compile	Completeness checksums, zero undetected errors, reference comparison, cycle time with review (31)
26	Medium	Roadmap	32 weeks, validation and UI last	Validation from week one, vertical slices, shadow mode (31)
27	Medium	Competition	Existing solutions = Veeva, MasterControl, OCR	Celegence, SyncIQ, Weave Bio, Peer AI, Veeva AI roadmap (5)
28	Medium	Regulation	No mention of AI-specific rules	FDA draft AI guidance (2025), FDA–EMA principles (2026), EU draft Annex 22 (18, 30)
29	Medium	Ledger	Hash chain alone makes the log immutable	Add external anchoring, insert-only role, entry signing (18)
30	Low	Business impact	Delay costs $1–3M per day	Average ~$0.8M/day (Tufts 2024); $1M+ only for large drugs (3)
31	Low	Business impact	IR or RTF halts the clock 30–90 days	IRs do not stop the clock; a major amendment adds 3 months (3)

#	Severity	Topic	Overview says	Fix (section)
32	Low	Regulation	Traceability comes from Part 11	Also 21 CFR 211 and FDA data-integrity guidance (4)
33	Low	Source systems	Empower has no API hooks	The Empower Toolkit exists; lab-data platforms use it (4)
34	Low	MCP	MCP replaces custom glue code	The glue moves inside each MCP server (11)
35	Low	Orchestration	LangGraph v0.3+	LangGraph 1.0 shipped Oct 2025 (14)
36	Low	Orchestration	Checkpoints in Redis	Postgres checkpoints, hashed into the ledger (14)
37	Low	Deployment	"Part 11 compliant" infrastructure	Compliance is the system plus procedures, validated by the customer (19)
38	Low	Deployment	Nitro Enclaves for isolation of inference	Enclaves suit CPU work such as keys, not GPU inference (19)
39	Low	Deployment	Tenant memory isolation in a single-tenant design	Pick one tenancy model and word it consistently (19)
40	Low	History	"LangChain 2021"	LangChain launched Oct 2022; LangGraph Jan 2024 (27)
41	Low	eCTD	3.2.P.5.5 for a stability narrative	Belongs in 3.2.P.8.3, supporting 3.2.P.5.6 (20)
42	Low	eCTD	Diagram shows 3 modules, no 3.2.R	5 modules; include 3.2.R regional (2)
43	Low	Numbers	$8.4B market, 12,000 hours, "60% scanned"	Unsourced; build bottom-up or cite (3, 32)

36. If you build it: a realistic student-scale version

You cannot get access to a real Veeva, LIMS, GovCloud or H100 cluster, and you do not need them. A focused version that proves the core principle (facts first, every number verified and traceable) is more impressive than a shallow copy of the whole enterprise diagram.

Scope: one fictional small-molecule tablet, two sections: 3.2.P.8.3 Stability Data and 3.2.P.5.4 Batch Analyses.

Data you create yourself:

- A synthetic **LIMS export** (CSV): 12 batches × 2 conditions (25°C/60% RH to 36 months, 40°C/75% RH to 6 months) × tests (assay, 2 named impurities, total impurities, dissolution), with realistic trends and noise.
- **Fake CMO Certificates of Analysis**: render some results as PDFs in three different layouts, then rasterize them and add blur, skew and noise so they look scanned. Because you generated them, you have perfect ground truth.
- **Injected errors**: transposed decimals (0.14 vs 0.014), unit swaps, a value that differs between LIMS and its CoA, an impossible date. These test the checker.
- **Retrieval corpus**: the public ICH guideline PDFs (M4Q, Q1A(R2), Q1E, Q3B(R2), Q6B).

Stack (free or cheap):

Layer	Choice	Why
Fact store + vectors + keyword search	PostgreSQL with pgvector and full-text search	One database, exact types, easy to explain
Lineage graph (optional)	Neo4j Community in Docker	Shows the graph idea for genealogy and impact queries
PDF / OCR	PyMuPDF for digital PDFs; Docling or PaddleOCR for scans	Word boxes for provenance
Vision + drafting model	Claude API (a Sonnet-class model for cost)	Vision, structured output, good writing
Workflow	LangGraph 1.x, Postgres checkpointer, an interrupt for human review	Shows durable state and human-in-the-loop

Layer	Choice	Why
Connectors	Two small MCP servers with the official Python SDK: "LIMS" over your CSV/Postgres, "DMS" over a folder of PDFs	Demonstrates MCP honestly
Statistics	A small library on NumPy, SciPy, statsmodels, with pytest tests and `Decimal` rounding	ICH Q1E shelf life, prediction-interval OOT
Review screen	React + PDF.js + Tiptap editor with fact chips	Click a number, see the box
Ledger	Postgres table with a SHA-256 hash chain and a verify script	Demo: edit a row by hand and watch verification fail

Metrics that make it convincing:

1. Extraction accuracy on your synthetic scans, and the share of wrong extractions that were **flagged** (target: all).
2. Share of injected errors caught by the validator.
3. Statistics matching an independent reference (statsmodels vs your library, or R).
4. Unverifiable numbers in final drafts: must be zero.
5. Tamper detection: the ledger check finds a manual edit.

Build order:

1. Synthetic data generator + fact schema.
2. LIMS path end to end: CSV → facts → statistics → stability table.
3. Drafting with fact placeholders + the number and claim validator.
4. Review screen with click-to-source.
5. Scanned CoA parsing with the OCR + VLM cross-check.
6. Ledger, LangGraph wiring, MCP servers.
7. Evaluation report with the five metrics.

How to pitch it in one breath: "An AI system for pharmaceutical regulatory dossiers where the language model never produces a number: every value comes from a verified fact store, is computed by tested code, links to its exact source cell, and is logged in a tamper-evident ledger. It even catches the science errors in the original design spec."

The whole-eCTD student version: three slices, one graph

Once the Module 3 slice above works, the right demo is not "all of eCTD" but **three thin slices from three modules, connected by the consistency engine** (section 8). That shows the moat in a way no single-module tool can.

Slice	Data (all public or synthetic)	What you build
Module 3: 3.2.P.8.3 stability	Synthetic LIMS export and fake CoAs (above)	Fact extraction, statistics library, stability table and text
Module 5: efficacy results	The R Consortium Pilot 1 submission package, public on GitHub: ADaM datasets (ADSL, ADAE, ADTTE and others) with define.xml, the analysis data reviewer's guide, R programs for the output tables, and a sample Module 1 cover letter	Read the output tables, re-derive headline numbers from ADaM, draft a CSR results paragraph and a 2.7.3 paragraph with placeholders
Label: sections 6 and 14	Your own drafted label for the fictional drug, formatted like real labels from the openFDA drug labeling API	Draft label sections 6 and 14 from the same facts; build the label claim matrix

The demo that sells it: change one number in the ADaM data (or plant a typo in the label) and show the platform (1) flag the mismatch between CSR, 2.7.3 and label, (2) list every affected placement, and (3) update the placeholders and queue the prose for re-review, with every step in the ledger. Then run the seeded-error study across all three slices and report the catch rate.

What to add to the stack above: a CDISC reader (pyreadstat or pandas for .xpt files, plus a define.xml parser), the evidence graph as Postgres tables plus Neo4j Community for placements and propagation, and one Agent Skill per document type (stability section, CSR results, label section).

37. Glossary

Every acronym in the two overview docs and this explainer, in plain words. Watch out for **API**, which means two different things in this project.

Term	Meaning
ALCOA+	Data-integrity standard: Attributable, Legible, Contemporaneous, Original, Accurate, plus Complete, Consistent, Enduring, Available
API (pharma)	Active Pharmaceutical Ingredient, the drug substance
API (software)	Application Programming Interface
BLA	Biologics License Application, to market a biologic in the US
BM25	Standard keyword-ranking formula used in search engines
CDER / CBER	FDA's drug center / biologics center (vaccines, blood, cell and gene therapy)
CDS	Chromatography Data System, such as Waters Empower
CFR	Code of Federal Regulations; Title 21 covers food and drugs
cGMP / GMP	(current) Good Manufacturing Practice, the legal quality rules for making medicines
CMC	Chemistry, Manufacturing and Controls
CMO / CDMO	Contract (Development and) Manufacturing Organization
CoA	Certificate of Analysis, one batch's test results against its specification
CRL	Complete Response Letter: FDA's "not approved yet, here is why"
CSV / CSA	Computer System Validation / FDA's risk-based Computer Software Assurance approach
DMS / RIM	Document Management System / Regulatory Information Management (Veeva Vault RIM)

Term	Meaning
eCTD	electronic Common Technical Document, the 5-module dossier format
ELN	Electronic Lab Notebook (Benchling)
EMA / PMDA / NMPA	Regulators of the EU / Japan / China
ERP / MES	Enterprise Resource Planning (SAP) / Manufacturing Execution System
ESG	FDA Electronic Submissions Gateway
FedRAMP	US government cloud security authorization program
FIPS 140-3	US government standard for validated cryptographic modules
GAMP 5	ISPE's industry guide to validating computerized GMP systems
GovCloud	AWS regions isolated for US government workloads
HPLC	High-Performance Liquid Chromatography, separates a sample into measurable peaks
HSM / KMS	Hardware Security Module / AWS Key Management Service, which keeps encryption keys
ICH	International Council for Harmonisation, writes the Q (quality) and M (format) guidelines
IND	Investigational New Drug application, to start human trials
IR	Information Request from FDA during review
JWT / OIDC / SAML	Login token format / modern single-sign-on protocol / older enterprise single-sign-on protocol
LIMS	Laboratory Information Management System
LOQ	Limit of Quantitation: below it a value is reported as "<LOQ"

Term	Meaning
MCP	Model Context Protocol, the standard plug between AI apps and tools
MSAT / TechOps	Manufacturing Science and Technology / Technical Operations
NDA	New Drug Application, to market a small-molecule drug in the US
NMT / NLT	Not More Than / Not Less Than, in specifications
OCR / VLM	Optical Character Recognition / Vision-Language Model
OOS / OOT	Out of Specification (broke a limit) / Out of Trend (unusual but within limits)
PDUFA	Law setting FDA review-time goals and user fees
QA / QC	Quality Assurance (audits the system) / Quality Control (runs the tests)
QOS	Quality Overall Summary, Module 2.3
RAG	Retrieval-Augmented Generation: search first, then let the model answer from what was found
RBAC	Role-Based Access Control
RH	Relative Humidity
RPA	Robotic Process Automation, bots that replay screen clicks
RTF	Refuse-to-File: FDA declines to review an incomplete application
TDI	Total Daily Intake of an impurity
vLLM	Open-source server for running language models fast on GPUs
VPC	Virtual Private Cloud, an isolated network in the cloud
WORM	Write Once, Read Many storage (S3 Object Lock)

38. Sources

Pages opened while checking the facts in this explainer, grouped by topic. The two overview docs in this project are the subject being explained.

Regulation and guidelines

- 21 CFR Part 11 (eCFR)
- 21 CFR 314.101, filing an NDA (eCFR)
- FDA draft guidance: Refuse to File, NDA and BLA submissions (2017)
- FDA review clock explained (Assyro)
- ICH Q1A(R2) Stability Testing
- ICH Q1E Evaluation of Stability Data
- ICH Q3B(R2) Impurities in New Drug Products
- New ICH Q1 draft guideline (ECA Academy)
- ICH revises Q1 guideline (Pharmaceutical Online)
- FDA eCTD requirements and v4.0 (IntuitionLabs)

AI regulation

- FDA draft guidance: AI to support regulatory decision-making (Jan 2025)
- DLA Piper summary of the FDA AI draft guidance
- FDA CDER: AI for drug development
- EMA and FDA align on Good AI Practice (Jones Day, Jan 2026)
- EU draft GMP Annex 22 on AI (PharmTech)
- FDA Computer Software Assurance final guidance (Federal Register, Sept 2025)

Models and AI infrastructure

- Claude Opus 4.6 launch coverage (R&D World)
- Anthropic models overview
- Anthropic model deprecations
- Amazon Bedrock models under FedRAMP / GovCloud (AWS)
- Claude in Amazon Bedrock approved for FedRAMP High (Anthropic, June 2025)
- DeepSeek V4 preview release
- States that banned DeepSeek (StateTech)

- MCP joins the Agentic AI Foundation
- LangChain and LangGraph 1.0
- Pinecone BYOC docs
- Tetra Empower Agent (TetraScience)
- Migrating off AWS QLDB (Certyo)

Market and competitors

- Tufts value of a day of delay (Contract Pharma)
- RIM system market (Grand View Research)
- Celegence automates CMC Module 3 (Manufacturing Chemist)
- AI CMC authoring software comparison 2026 (IntuitionLabs)
- Veeva unveils Falcon and Agentic Authoring (Clinical Trial Vanguard)

GenAI and agentic technologies (section 28)

- GEPA: Reflective Prompt Evolution (arXiv, ICLR 2026)
- DSPy GEPA optimizer docs
- V-JEPA 2 (Meta, GitHub)
- LLM-JEPA (arXiv)
- AMI Labs and JEPA (TorontoStarts)
- Claude structured outputs
- Claude citations
- Open-source OCR and document VLMs compared (Spheron)
- ColPali (Hugging Face)
- MCP 2025-11-25 spec update (WorkOS)
- MCP 2026-07-28 release candidate
- Advanced tool use (Anthropic Engineering)
- Agent Skills open standard (SiliconANGLE)
- State of OpenTelemetry GenAI conventions, July 2026
- A2A protocol one year on (Linux Foundation)

Whole-eCTD platform (sections 7–9, 21–26, 29, 33)

- Weave Bio and Takeda across 14 programs (HIT Consultant, Sept 2026)
- Weave Bio launches NDA workflow with Parexel (Business Wire, April 2026)
- FDA binding guidance on standardized study data (CDISC)

- FDA Study Data Technical Conformance Guide
- eCTD v4.0 implementation timelines (IntuitionLabs)
- ICH M11 in force (Sakara Digital)
- ICH M4Q(R2) changes to Modules 2.3 and 3 (EUREG)
- R Consortium Pilot 1 submission package (GitHub)
- openFDA drug labeling API
- FDA's 202 published CRLs, 74% citing quality/manufacturing (Pharma Manufacturing)

Weave Bio product and design (section 34)

- Weave Bio home page
- Weave Bio Submission Builder
- Weave Bio HAQ Manager
- Weave Bio security and privacy
- About Weave Bio
- AutoIND study by Weave and Takeda authors (arXiv 2509.09738)
- Regfo vs Weave Bio, written by Regfo's founder (BrightByte)
- Weave Bio company profile (Excedr)